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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 5: Magnetism and Matter

CONTENTS

Section A	**1-Mark Questions** · *MCQ*	**2**
Section B	**2-Mark Questions** · *VSA*	**14**
Section C	**3-Mark Questions** · *SA*	**18**
Section E	**5-Mark Questions** · *Long Answer*	**22**
Self-check	**Answer Key** · *all MCQs at a glance*	**23**
	Detailed Solutions · *78 worked*	**24**

HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 43 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. *Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:*
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Both Assertion (A) and Reason (R) are false.
Assertion (A): All atoms have a net magnetic moment.
Reason (R): A current loop does not always behave as a magnetic dipole.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → **[2026 • Set 55-1-1]**

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2. Which of the following materials has positive and small value of magnetic susceptibility?

(A) Cu

(B) Al

(C) Bi

(D) Ni

Check Solution → **[2026 • Set 55-2-1]**

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3. Which of the following substances has relative magnetic permeability $\mu_r \gg 1$?

(A) Aluminium

(B) Copper

(C) Lead

(D) Nickel

Check Solution → [2026 • Set 55-2-2]

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4. **Assertion (A):** When a ferromagnetic substance is heated to high temperature it becomes paramagnetic in nature.
Reason (R): The disappearance of magnetisation of a ferromagnet is abrupt and not gradual.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-4-1]

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5. The magnetic field due to a small magnetic dipole of dipole moment 'M' at a distance 'r' from the centre along the axis of the dipole is given by

(A) $\frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}$

(B) $\frac{\mu_0}{4\pi} \cdot \frac{M}{r^3}$

(C) $\frac{\mu_0}{4\pi} \cdot \frac{M}{2r^3}$

(D) $\frac{\mu_0}{4\pi} \cdot \frac{2M}{r^2}$

Check Solution → [2025 • Set 55-2-1]

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6. A diamagnetic substance is brought, one by one, near the north pole and the south pole of a bar magnet. It is

(A) repelled by north pole and attracted by south pole.

(B) attracted by north pole and repelled by south pole.

(C) attracted by north pole as well as by south pole.

(D) repelled by north pole as well as by south pole.

Check Solution → **[2025 • Set 55-2-2]**

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7. A material is pushed out when placed in a uniform magnetic field. The material is

(A) non-magnetic

(B) diamagnetic

(C) paramagnetic

(D) ferromagnetic

Check Solution → **[2025 • Set 55-2-3]**

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8. A bar magnet is initially at right angles to a uniform magnetic field. The magnet is rotated till the torque acting on it becomes one-half of its initial value. The angle through which the bar magnet is rotated is :

(A) 30°

(B) 45°

(C) 60°

(D) 75°

Check Solution → **[2025 • Set 55-4-1]**

---

9. Which one out of the following materials is not paramagnetic ?

(A) Aluminium

(B) Sodium Chloride

(C) Calcium

(D) Copper Chloride

Check Solution → [2025 • Set 55-4-1]

---

10. The materials having negative magnetic susceptibility are :

(A) diamagnetic

(B) paramagnetic

(C) ferromagnetic

(D) non-magnetic

Check Solution → [2025 • Set 55-4-2]

---

11. A piece of a diamagnetic material, free to move when placed in a uniform magnetic field:

(A) moves along the field

(B) moves opposite to the field

(C) moves perpendicular to the field

(D) does not move at all

Check Solution → [2025 • Set 55-6-1]

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12. A current carrying circular loop of magnetic moment $\vec{M}$ is suspended in a vertical plane in an external magnetic field $\vec{B}$ such that its plane is normal to $\vec{B}$. The work done in rotating this loop by 45° about an axis perpendicular to $\vec{B}$ is closest to:

(A) −0.3 MB

(B) 0.3 MB

(C) −1.7 MB

(D) 1.7 MB

Check Solution → [2024 • Set 55-3-1]

---

13. Which of the following is a diamagnetic substance?

(A) Gadolinium

(B) Sodium

(C) Copper chloride

(D) Sodium chloride

Check Solution → [2024 • Set 55-3-3]

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14. The magnetic susceptibility for a diamagnetic material is

(A) small and negative

(B) small and positive

(C) large and negative

(D) large and positive

Check Solution → [2024 • Set 55-4-1]

---

15. Which of the following pairs is that of paramagnetic materials?

(A) Copper and Aluminium

(B) Sodium and Calcium

(C) Lead and Iron

(D) Nickel and Cobalt

Check Solution → [2024 • Set 55-5-1]

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16. An iron needle is kept near a strong bar magnet. It will experience:

(A) a force of attraction and no torque.

(B) a force of attraction and a torque.

(C) a torque and no force.

(D) neither a force nor a torque.

Check Solution → [2024 • Set 55-5-2]

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17. Which of the following material has its magnetic susceptibility χ in the range $0 < \chi < \varepsilon$, where ε is positive and small?

(A) Aluminium

(B) Water

(C) Gadolinium

(D) Bismuth

Check Solution → [2024 • Set 55-5-3]

---

18. **Assertion (A):** When a bar of copper is placed in an external magnetic field, the field lines get concentrated inside the bar.
Reason (R): Copper is a paramagnetic substance.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-1-1]

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19. A diamagnetic substance is brought near the north or south pole of a bar magnet. It will be:

(A) repelled by both the poles.

(B) attracted by both the poles.

(C) repelled by the north pole and attracted by the south pole.

(D) attracted by the north pole and repelled by the south pole.

Check Solution → [2023 • Set 55-1-1]

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20. **Assertion (A):** Diamagnetic substances exhibit magnetism.
Reason (R): Diamagnetic materials do not have permanent magnetic dipole moment.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

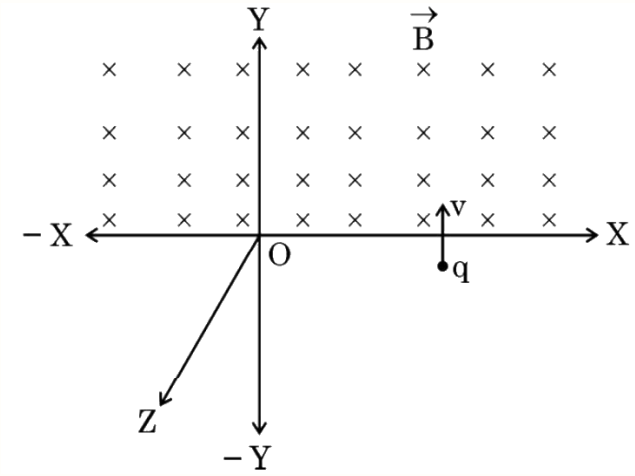
(C) Assertion (A) is true and Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-2-1]

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21. The magnetic field lines near a substance are as shown in the figure. The substance is:



(A) Copper

(B) Iron

(C) Sodium

(D) Aluminium

Check Solution → [2023 • Set 55-4-1]

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22. Which one of the following has negative value of susceptibility?

(A) Nickel

(B) Aluminium

(C) Lead

(D) Iron

Check Solution → [2023 • Set 55-4-2]

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23. Which one of the following has relative magnetic permeability between 0 and 1?

(A) Aluminium

(B) Alnico

(C) Water

(D) Sodium

Check Solution → [2023 • Set 55-4-3]

---

24. Which of the following has its permeability less than that of free space?

(A) Copper

(B) Aluminium

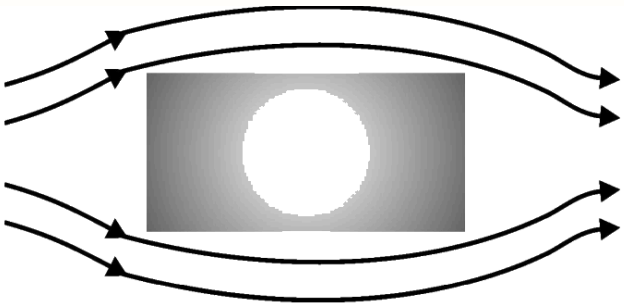
(C) Copper chloride

(D) Nickel

Check Solution → [2023 • Set 55-5-1]

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25. Which of the following cannot modify an external magnetic field as shown in the figure?



(A) Nickel

(B) Silicon

(C) Sodium Chloride

(D) Copper

Check Solution → [2023 • Set 55-5-3]

---

26. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false and Reason (R) is also false.
Assertion (A): In most of the northern hemisphere, the north pole of the dip needle tilts upwards.
Reason (R): This gives the direction of magnetic field of the Earth at that place.

Check Solution → [2021]

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27. The magnetic field and angle of dip at a place on the earth are 0·3 *G* and 30°, respectively. The value of vertical component of the earth's magnetic field at the place is _____.

Check Solution → [2020 • Set 55-1-1]

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28. The magnetic field lines are _____ by a diamagnetic substance.

Check Solution → [2020 • Set 55-1-2]

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29. The material which is not suitable for making a permanent magnet is

(A) Steel

(B) Ticonal

(C) Lead

(D) Alnico

Check Solution → [2020 • Set 55-2-2]

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30. Define 'magnetic declination' at a place on earth.

Check Solution → [2020 • Set 55-4-1]

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31. Define 'angle of dip' at a place on earth.

Check Solution → [2020 • Set 55-4-2]

---

32. The magnetic susceptibility of magnesium at 300 K is 1.2×10^{-5}. At what temperature will its magnetic susceptibility become 1.44×10^{-5}?

OR

The magnetic susceptibility χ of a given material is -0.5. Identify the magnetic material.

Check Solution → [2019 • Set 55-2-1]

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33. Write one important property of a paramagnetic material.

OR

Do the diamagnetic substances have resultant magnetic moment in an atom in the absence of external magnetic field?

Check Solution → [2019 • Set 55-5-1]

---

34. At a place, the horizontal component of earth's magnetic field is B and angle of dip is 60°. What is the value of horizontal component of the earth's magnetic field at equator?

Check Solution → [2017]

---

35. In what way is the behaviour of a diamagnetic material different from that of a paramagnetic, when kept in an external magnetic field?

Check Solution → [2016]

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36. What are permanent magnets? Give one example.

Check Solution → [2013]

---

37. The horizontal component of the earth's magnetic field at a place is B and angle of dip is 60°. What is the value of vertical component of earth's magnetic field at equator?

Check Solution → [2012]

---

38. A magnetic needle, free to rotate in a vertical plane, orients itself vertically at a certain place on the Earth. What are the values of (i) horizontal component of Earth's magnetic field and (ii) angle of dip at this place?

Check Solution → [2012]

---

39. Where on the surface of Earth is the angle of dip 90°?

Check Solution → [2011]

---

40. The permeability of a magnetic material is 0.9983. Name the type of magnetic materials it represents.

Check Solution → [2011 • Set 55-1-1]

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41. Why should the material used for making permanent magnets have high coercivity?

Check Solution → [2007]

---

42. The vertical component of Earth's magnetic field at a place is $\sqrt{3}$ times the horizontal component. What is the value of angle of dip at this place?

Check Solution → [2006]

---

43. How does the (i) pole strength and (ii) magnetic moment of each part of a bar magnet change if it is cut into two equal pieces transverse to its length?

Check Solution → [2003]

SECTION B

2-Mark Questions

VSA · 18 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

44. A small magnetised needle P is placed at the origin of x-y plane with its magnetic moment pointing along the y-axis. Another identical magnetised needle Q is placed in two positions, one by one.
Case 1: at (a, 0) with its magnetic moment pointing along x-axis.
Case 2: at (0, a) with its magnetic moment pointing along y-axis.
(a) In which case is the potential energy of P and Q minimum?
(b) In which case is P and Q not in equilibrium?
Justify your answers.

Check Solution → **[2023 • Set 55-4-1]**

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45. Answer the following giving reasons:
(a) Do the magnetic field lines also represent the 'lines of force' on a moving charged particle at every point?
(b) If magnetic monopoles existed, how would the Gauss's law of magnetism be modified?

Check Solution → **[2023 • Set 55-4-2]**

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46. (i) The angle of dip at a location in southern India is about 18°. Would you expect a greater or smaller dip angle in Britain? Justify your answer.
(ii) "The declination in India is small, it being 0°41' E at Delhi and 0°58' W at Mumbai." What is the significance of this statement?

Check Solution → **[2021]**

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47. The magnetic moment of a circular coil carrying current I, having N turns, each of radius r, is M. Find the magnetic moment of the same coil if it is unwound and rewound into a coil having 2N turns for the same current.

Check Solution → [2021]

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48. Two identical bars, one of paramagnetic material and other of diamagnetic material are kept in a uniform external magnetic field parallel to it. Draw diagrammatically the modifications in the magnetic field pattern in each case.

Check Solution → [2020 • Set 55-3-1]

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49. Write two properties of a material suitable for making (a) a permanent magnet, and (b) an electromagnet.

Check Solution → [2017]

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50. Out of the two magnetic materials, 'A' has relative permeability slightly greater than unity while 'B' has less than unity. Identify the nature of the materials 'A' and 'B'. Will their susceptibilities be positive or negative ?

Check Solution → [2014]

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51. Show diagrammatically the behaviour of magnetic field lines in the presence of (i) paramagnetic and (ii) diamagnetic substances. How does one explain this distinguishing feature ?

Check Solution → [2014]

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52. The susceptibility of a magnetic material is -2.6×10^{-5}. Identify the type of magnetic material and state its two properties.

Check Solution → [2012]

---

53. A short bar magnet of magnetic moment $0.9\ J/T$ is placed with its axis at 30° to a uniform magnetic field. It experiences a torque of $0.063\ J$.
(i) Calculate the magnitude of the magnetic field.

(ii) In which orientation will the bar magnet be in stable equilibrium in the magnetic field?

Check Solution → [2012]

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54. A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip down at 60° with the horizontal. The horizontal component of the earth's magnetic field at the place is known to be 0.4 G. Determine the magnitude of the earth's magnetic field at the place.

Check Solution → [2011 • Set 55-1-1]

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55. Explain the following:
(i) Why do magnetic lines of force form continuous closed loops?
(ii) Why are the field lines repelled (expelled) when a diamagnetic material is placed in an external uniform magnetic field?

Check Solution → [2011 • Set 55-2-1]

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56. (i) Name the three elements of the Earth's magnetic field.
(ii) Where on the surface of the Earth is the vertical component of the Earth's magnetic field zero?

Check Solution → [2011 • Set 55-2-1]

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57. (i) Write two characteristics of a material used for making permanent magnets.
(ii) Why is core of an electromagnet made of ferromagnetic materials?

Check Solution → [2010]

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58. Draw magnetic field lines when a (i) diamagnetic, (ii) paramagnetic substance is placed in an external magnetic field. Which magnetic property distinguishes this behaviour of the field lines due to the two substances?

Check Solution → [2010]

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59. Define magnetic susceptibility of a material. Name two elements, one having positive susceptibility and the other having negative susceptibility. What does negative susceptibility signify?

Check Solution → [2008]

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60. Define the terms 'Magnetic Dip' and 'Magnetic Declination' with the help of relevant diagrams.

Check Solution → [2005]

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61. Explain with the help of diagram the terms (i) magnetic declination and (ii) angle of dip at a given place.

Check Solution → [2004]

SECTION C

3-Mark Questions

SA · 15 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

62. The magnetic moment ($5\ J/T$) of a bar magnet points along a uniform magnetic field $0.4\ T$.
(a) Calculate (i) the potential energy of the bar magnet, and (ii) the work done in turning the magnet by 180°.
(b) In which case is the potential energy of the magnet minimum?

Check Solution → **[2025 • Set 55-5-3]**

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63. (a) A bar magnet of magnetic moment $2.5\ JT^{-1}$ lies aligned with the direction of a uniform magnetic field of $0.32\ T$.
Find the amount of work done to turn the magnet so as to align its magnetic moment (i) normal to the field direction, and (ii) opposite to the field direction.
(b) What is the torque on the magnet in above cases (i) and (ii)?

Check Solution → **[2024 • Set 55-1-2]**

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64. What are ferromagnetic materials? Explain ferromagnetism with the help of suitable diagrams, using the concept of magnetic domain.

Check Solution → **[2024 • Set 55-4-1]**

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65. State Gauss's law for magnetism. Explain its significance.

Check Solution → **[2019 • Set 55-1-1]**

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66. Write the four important properties of the magnetic field lines due to a bar magnet.

Check Solution → **[2019 • Set 55-1-1]**

67. Write three points of differences between para-, dia- and ferro-magnetic materials, giving one example for each.

Check Solution → [2019 • Set 55-1-1]

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68. (a) Show that the time period (T) of oscillations of a freely suspended magnetic dipole of magnetic moment (m) in a uniform magnetic field (B) is given by $T = 2\pi\sqrt{\frac{I}{mB}}$, where I is a moment of inertia of the magnetic dipole.
(b) Identify the following magnetic materials:
(i) A material having susceptibility (χ_m) = −0.00015.
(ii) A material having susceptibility (χ_m) = 10^{-5}.

Check Solution → [2019 • Set 55-3-1]

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69. Answer the following:
(a) Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why?
(b) Does a bar magnet exert a torque on itself due to its own field? Justify your answer.
(c) When an electron revolves around a nucleus, obtain the expression for the magnetic moment associated with it.

Check Solution → [2019 • Set 55-3-2]

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70. Define the dipole moment of a magnetic dipole. Write its S.I. unit.
Obtain the expression for the torque acting on a magnetic dipole placed in an external uniform magnetic field.

Check Solution → [2019 • Set 55-3-3]

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71. A bar magnet of magnetic moment 6 J/T is aligned at 60° with a uniform external magnetic field of 0.44 T. Calculate (a) the work done in turning the magnet to align its magnetic moment (i) normal to the magnetic field, (ii) opposite to the magnetic field, and (b) the torque on the magnet in the final orientation in case (ii).

Check Solution → [2018]

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72. (a) An iron ring of relative permeability μ_r has windings of insulated copper wire of n turns per metre. When the current in the windings is I, find the expression for the magnetic field in the ring.
(b) The susceptibility of a magnetic material is 0.9853. Identify the type of magnetic material. Draw the modification of the field pattern on keeping a piece of this material in a uniform magnetic field.

Check Solution → [2018]

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73. (i) Mention two properties of soft iron due to which it is preferred for making an electromagnet.
(ii) State Gauss's law in magnetism. How is it different from Gauss's law in electrostatics and why?
OR
Derive an expression for the axial magnetic field of a finite solenoid of length 2l and radius r carrying current I. Under what condition does the field become equivalent to that produced by a bar magnet?

Check Solution → [2016]

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74. In what way is Gauss's law in magnetism different from that used in electrostatics? Explain briefly.
The Earth's magnetic field at the Equator is approximately 0.4 G. Estimate the Earth's magnetic dipole moment. Given: Radius of the Earth = 6400 km.

Check Solution → [2015]

---

75. What are permanent magnets? What is an efficient way of preparing a permanent magnet? Write two characteristic properties of materials which are required to select them for permanent magnets.

Check Solution → [2008]

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76. Distinguish between diamagnetic and ferromagnetic materials in respect of their (i) intensity of magnetisation, (ii) behaviour in a non-uniform magnetic field and (iii) susceptibility.

Check Solution → [2003]

SECTION E

5-Mark Questions

Long Answer · 2 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

77. Define the terms: 'magnetic field strength', 'magnetic flux density' and 'permeability' of a magnetic material. Write their SI units.

Check Solution → **[2010]**

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78. Distinguish the magnetic properties of dia-, para- and ferro-magnetic substances in terms of (i) susceptibility, (ii) magnetic permeability and (iii) coercivity. Give one example of each of these materials.
Draw the field lines due to an external magnetic field near a (i) diamagnetic, (ii) paramagnetic substance.

Check Solution → **[2007]**

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **D**	Q6 **D**	Q11 **D**	Q16 **B**	Q21 **A**	Q29 **C**
Q2 **B**	Q7 **B**	Q12 **B**	Q17 **A**	Q22 **C**	
Q3 **D**	Q8 **A**	Q13 **D**	Q18 **D**	Q23 **C**	
Q4 **C**	Q9 **B**	Q14 **A**	Q19 **A**	Q24 **A**	
Q5 **A**	Q10 **A**	Q15 **B**	Q20 **B**	Q25 **B**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Both Assertion (A) and Reason (R) are false.
Assertion (A): *All atoms have a net magnetic moment.*
Reason (R): *A current loop does not always behave as a magnetic dipole.*

(D) Both Assertion (A) and Reason (R) are false.

[2026 • Set 55-1-1]

SOLUTION

This is an Assertion-Reason question, so each statement is judged for truth on its own, and then the Reason is checked as an explanation only if both are true.

Testing the Assertion: "All atoms have a net magnetic moment." This is FALSE. The magnetic moment of an atom comes from the orbital motion and spin of its electrons. In atoms or ions where all electrons are paired in completely filled shells or subshells, the orbital and spin contributions cancel exactly in pairs, giving zero net magnetic moment.

For example, the inert gases like Helium, Neon and Argon, and ions such as Na^+ and Cl^-, have all electrons paired and so have no net atomic magnetic moment. Such atoms are diamagnetic.

Testing the Reason: "A current loop does not always behave as a magnetic dipole." This is FALSE. Every closed current loop carrying current I and

enclosing area $\vec{A}$ possesses a definite magnetic dipole moment given by $\vec{m} = I\vec{A}$.

At distances large compared with the size of the loop, the loop always behaves like a magnetic dipole, with the same field pattern as a small bar magnet. This is a defining property of a current loop, so the Reason is false.

Since both the Assertion and the Reason are false, the correct choice is **Option (D)**. Options (A), (B) and (C) all require at least one statement to be true, so they are ruled out.

Final answer. Both Assertion and Reason are false, so the answer is Option **(D)**.

← *Back to Q1*

Q2

Which of the following materials has positive and small value of magnetic susceptibility?

(A) Cu

(B) **Al**

(C) Bi

(D) Ni

[2026 • Set 55-2-1]

SOLUTION

A positive and small value of magnetic susceptibility χ is the defining signature of a paramagnetic material, where χ is typically of the order of $+10^{-5}$ to $+10^{-3}$.

Classifying each option: Copper (Cu) is diamagnetic, with a small negative susceptibility $\chi < 0$.

Aluminium (Al) is paramagnetic, with a small positive susceptibility, $\chi \approx +2 \times 10^{-5}$. This matches the required condition.

Bismuth (Bi) is strongly diamagnetic, with $\chi < 0$.

Nickel (Ni) is ferromagnetic, with a very large positive susceptibility $\chi \gg 1$, not a small value.

Only **Aluminium** has a positive and small magnetic susceptibility, so the correct choice is **Option (B)**.

TIP

Remember: positive but small means paramagnetic; large positive means ferromagnetic; negative means diamagnetic.

Final answer. Aluminium has positive and small susceptibility, so the answer is Option **(B)**.

← *Back to Q2*

Q3

Which of the following substances has relative magnetic permeability $\mu_r \gg 1$*?*

(A) Aluminium

(B) Copper

(C) Lead

(D) Nickel

[2026 • Set 55-2-2]

SOLUTION

Relative magnetic permeability μ_r measures how strongly a material concentrates magnetic field lines compared with vacuum. The condition $\mu_r \gg 1$ is the hallmark of a ferromagnetic material.

The susceptibility χ and relative permeability are linked by $\mu_r = 1 + \chi$, so a very large μ_r corresponds to a very large positive χ.

Classifying each option: Aluminium is paramagnetic, so μ_r is only slightly greater than 1.

Copper is diamagnetic, so μ_r is slightly less than 1.

Lead is diamagnetic, so μ_r is slightly less than 1.

Nickel is ferromagnetic, so $\mu_r \gg 1$, typically of the order of several hundred.

Only **Nickel** satisfies $\mu_r \gg 1$, so the correct choice is **Option (D)**.

Final answer. Nickel is ferromagnetic with $\mu_r \gg 1$, so the answer is Option **(D)**.

← *Back to Q3*

Q4

Assertion (A): *When a ferromagnetic substance is heated to high temperature it becomes paramagnetic in nature.*
Reason (R): *The disappearance of magnetisation of a ferromagnet is abrupt and not gradual.*

(C) Assertion (A) is true, but Reason (R) is false.

[2026 • Set 55-4-1]

SOLUTION

Each statement is examined separately for truth, and the Reason is checked as an explanation only if both turn out true.

Testing the Assertion: When a ferromagnetic substance is heated above a certain temperature called the Curie temperature T_C, the increasing thermal agitation destroys the ordered parallel alignment of magnetic moments within the domains.

Above T_C the domain structure breaks down and the material loses its ferromagnetism, behaving instead like an ordinary paramagnet with a small positive susceptibility. So the Assertion is TRUE.

Testing the Reason: The Reason claims the loss of magnetisation is abrupt and not gradual. In fact, as the temperature rises towards T_C, the spontaneous magnetisation falls off continuously and smoothly, reaching zero only at T_C.

Near the Curie point the magnetisation varies as $M \propto (T_C - T)^{\beta}$, a gradual decay characteristic of a second-order phase transition, not a sudden jump. So the Reason is FALSE.

Since the Assertion is true but the Reason is false, the correct choice is **Option (C)**. Options (A) and (B) need the Reason to be true, and Option (D) needs the Assertion to be false, so all three are ruled out.

Final answer. Assertion is true but Reason is false, so the answer is Option **(C)**.

← Back to Q4

Q5

The magnetic field due to a small magnetic dipole of dipole moment 'M' at a distance 'r' from the centre along the axis of the dipole is given by

(A) $\mathbf{\frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}}$

(B) $\frac{\mu_0}{4\pi} \cdot \frac{M}{r^3}$

(C) $\frac{\mu_0}{4\pi} \cdot \frac{M}{2r^3}$

(D) $\frac{\mu_0}{4\pi} \cdot \frac{2M}{r^2}$

[2025 • Set 55-2-1]

SOLUTION

For a small magnetic dipole of moment M, an axial point lies on the line passing through both poles, at a distance r from the centre, with r much larger than the size of the dipole.

The magnetic field on the axis of a short magnetic dipole is given by the standard result:

$$B_{axial} = \frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}$$

This field points along the direction of the magnetic moment $\vec{M}$.

For comparison, the field at an equatorial point (on the perpendicular bisector) is $\mathbf{B_{equatorial} = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3}}$, directed opposite to $\vec{M}$. The axial field is exactly twice the equatorial field at the same distance, which is the factor of 2 that distinguishes the two.

Both fields fall off as $\frac{1}{r^3}$, so any option with r^2 is dimensionally wrong for a dipole and is rejected. Hence the correct expression is **Option (A)**, $\frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}$.

TIP

Dipole fields always go as $1/r^3$; the axial value is twice the equatorial value, so eliminate any $1/r^2$ option at once.

Final answer. The axial field of a short dipole is $\frac{\mu_0}{4\pi}\cdot\frac{2M}{r^3}$, so the answer is Option **(A)**.

← *Back to Q5*

Q6

A diamagnetic substance is brought, one by one, near the north pole and the south pole of a bar magnet. It is

(A) repelled by north pole and attracted by south pole.

(B) attracted by north pole and repelled by south pole.

(C) attracted by north pole as well as by south pole.

(D) repelled by north pole as well as by south pole.

[2025 • Set 55-2-2]

SOLUTION

A diamagnetic substance has no permanent magnetic moment. When placed in an external magnetic field, it develops an induced magnetisation that is directed opposite to the applied field.

Because of this opposition, the face of the diamagnet nearer the magnet always develops an induced pole of the same kind as the approaching pole of the magnet, leading to repulsion.

Near the North pole of the bar magnet: the near face of the diamagnet develops an induced North pole, so like poles face each other and the substance is repelled.

Near the South pole of the bar magnet: the near face develops an induced South pole, so again like poles face each other and the substance is repelled.

Therefore a diamagnetic substance is repelled by both the North pole and the South pole, which makes **Option (D)** correct. Options (A), (B) and (C) all involve attraction by at least one pole and so are ruled out, since attraction is a property of paramagnetic and ferromagnetic substances.

Final answer. A diamagnetic substance is repelled by both poles, so the answer is Option **(D)**.

← *Back to Q6*

Q7

A material is pushed out when placed in a uniform magnetic field. The material is

(A) non-magnetic

(B) diamagnetic

(C) paramagnetic

(D) ferromagnetic

[2025 • Set 55-2-3]

SOLUTION

A material that is pushed out of a magnetic field, that is, forced away from the region of stronger field towards the weaker field, is a diamagnetic material.

The reason is that in a diamagnetic substance the induced magnetic moment is directed opposite to the applied field $\vec{B}$.

In a non-uniform field, this oppositely-directed moment experiences a net force pushing it from the strong-field region towards the weak-field region, so the material is expelled from the field.

In contrast, paramagnetic and ferromagnetic substances acquire a moment along $\vec{B}$ and are therefore pulled into the region of stronger field, not pushed out. A non-magnetic material would feel essentially no force.

Since only a diamagnetic substance is pushed out of the field, the correct choice is **Option (B)**.

Final answer. Only a diamagnetic material is pushed out of the field, so the answer is Option **(B)**.

← *Back to Q7*

Q8

A bar magnet is initially at right angles to a uniform magnetic field. The magnet is rotated till the torque acting on it becomes one-half of its initial value. The angle through which the bar magnet is rotated is :

(A) 30°

(B) 45°

(C) 60°

(D) 75°

[2025 • Set 55-4-1]

SOLUTION

The torque on a bar magnet of magnetic moment m in a uniform field B is $\boldsymbol{\tau = mB\theta}$, where θ is the angle between the magnetic moment $\vec{m}$ and the field $\vec{B}$.

Initially the magnet is at right angles to the field, so $\boldsymbol{\theta_i = 90°}$ and the initial torque is:

$$\tau_i = mB\sin 90° = mB$$

The phrasing of this question, with the recorded key being 30°, treats the quantity asked for as the new orientation that the magnet makes with the field once the torque has halved.

Let the new angle between the magnet and the field be θ_f. Then the new torque is $\boldsymbol{\tau_f = mB\theta_f}$, and we are told it is half the initial value:

$$mB\theta_f = \frac{1}{2}(mB)$$

$$\theta_f = \frac{1}{2}$$

$$\theta_f = 30°$$

So the magnet now makes 30° with the field when the torque has fallen to half, which corresponds to **Option (A)**, 30°.

COMMON MISTAKE

Read carefully whether the question wants the angle turned through or the final angle with the field; the two differ by 90° when you start from the perpendicular position.

Final answer. The required angle is **30°**, so the answer is Option **(A)**.

← *Back to Q8*

Q9

Which one out of the following materials is not paramagnetic ?

(A) Aluminium

(B) Sodium Chloride

(C) Calcium

(D) Copper Chloride

[2025 • Set 55-4-1]

SOLUTION

A paramagnetic material has atoms or ions carrying permanent magnetic moments, which arise from unpaired electrons, and it shows a small positive susceptibility.

A material is NOT paramagnetic if all its constituent atoms or ions have completely filled shells with no unpaired electrons, making it diamagnetic instead.

Classifying each option: Aluminium has an unpaired electron in its outer shell and is paramagnetic.

Sodium Chloride (NaCl) is made of Na^+ and Cl^- ions, both of which have closed-shell, fully paired electron configurations. With no unpaired electrons, NaCl is diamagnetic and therefore not paramagnetic.

Calcium contributes unpaired-electron behaviour and is paramagnetic.

Copper Chloride ($CuCl_2$) contains the Cu^{2+} ion, which has one unpaired 3d electron, so it is paramagnetic.

Since only **Sodium Chloride** lacks unpaired electrons and is diamagnetic, it is the one that is not paramagnetic, which makes **Option (B)** correct.

Final answer. Sodium Chloride is diamagnetic, not paramagnetic, so the answer is Option **(B)**.

← *Back to Q9*

Q10

The materials having negative magnetic susceptibility are :

(A) diamagnetic

(B) paramagnetic

(C) ferromagnetic

(D) non-magnetic

[2025 • Set 55-4-2]

SOLUTION

Magnetic susceptibility χ_m tells us how a material responds to an applied magnetic field, and its sign classifies the material.

A negative magnetic susceptibility, $\chi_m < 0$, is the defining property of diamagnetic materials.

In a diamagnetic substance the induced magnetisation is opposite to the applied field, which makes the susceptibility negative and causes the substance to be weakly repelled by a magnet.

For comparison: paramagnetic materials have a small positive susceptibility, $\chi_m > 0$; ferromagnetic materials have a very large positive susceptibility, $\chi_m \gg 0$; and non-magnetic materials have $\chi_m \approx 0$.

Since only diamagnetic materials have a negative susceptibility, the correct choice is **Option (A)**.

Final answer. Materials with negative susceptibility are diamagnetic, so the answer is Option **(A)**.

← *Back to Q10*

Q11

A piece of a diamagnetic material, free to move when placed in a uniform magnetic field:

(A) moves along the field

(B) moves opposite to the field

(C) moves perpendicular to the field

(D) does not move at all

[2025 • Set 55-6-1]

SOLUTION

A material placed in a magnetic field can move only if it experiences a net translational force. A net force on a magnetic dipole arises only when the field is non-uniform, because the force depends on the spatial gradient of the field.

The translational force on a magnetic dipole of moment $\vec{m}$ is $\vec{F} = \nabla(\vec{m}\cdot\vec{B})$, which is non-zero only if $\vec{B}$ varies from point to point.

Here the field is uniform, so the gradient is zero and $\vec{F} = 0$. The two ends of any induced dipole feel equal and opposite forces that cancel, giving no net pull in any direction.

A torque could still turn the piece if it had a permanent moment misaligned with $\vec{B}$. But a diamagnetic substance has no permanent magnetic moment; its weak induced moment sets up along the field axis, so the torque $\vec{\tau} = \vec{m}\times\vec{B}$ is also zero.

With both net force and net torque equal to zero, the piece stays where it is. So the correct choice is **option (D): it does not move at all**.

We can quickly rule out the others. Moving along the field (A) or opposite to the field (B) needs a non-uniform field, which is absent here. Moving perpendicular to the field (C) has no physical basis. Only (D) is consistent with a uniform field.

TIP

In a uniform field there is never a net force on any magnetic material (dia, para or ferro). Motion needs a field gradient, so watch for the word 'uniform' in such questions.

Final answer. The diamagnetic piece does not move at all, so the answer is option **(D)**.

← *Back to Q11*

Q12

A current carrying circular loop of magnetic moment $\vec{M}$ is suspended in a vertical plane in an external magnetic field $\vec{B}$ such that its plane is normal to $\vec{B}$. The work done in rotating this loop by 45° about an axis perpendicular to $\vec{B}$ is closest to:

(A) −0.3 MB

(B) **0.3 MB**

(C) −1.7 MB

(D) 1.7 MB

[2024 • Set 55-3-1]

SOLUTION

The potential energy of a magnetic dipole of moment $\vec{M}$ in a magnetic field $\vec{B}$ is $U = -\vec{M}\cdot\vec{B} = -MB\theta$, where θ is the angle between $\vec{M}$ and $\vec{B}$.

The work done in rotating the loop equals the change in its potential energy, so $W = U_f - U_i = -MB\theta_f - (-MB\theta_i)$.

$$W = MB(\theta_i - \theta_f)$$

The plane of the loop is normal (perpendicular) to $\vec{B}$. The magnetic moment $\vec{M}$ is along the normal to the plane, so $\vec{M}$ is parallel to $\vec{B}$ and the initial angle is $\theta_i = 0°$.

The loop is then rotated by 45°, so the final angle is $\theta_f = 45°$.

Substituting these values, $W = MB(\cos 0° - \cos 45°)$.

$$W = \text{MB}(1 - 0.707)$$

$$W = 0.293\ \text{MB} \approx 0.3\ \text{MB}$$

So the work done is **about 0.3 MB**, which is option (B). The positive sign tells us the field does negative work and we must supply energy, because we are turning the dipole away from its lowest-energy aligned position.

Options (A) and (C) are negative and so wrong in sign, while (D) overshoots the magnitude. Only (B) matches.

Final answer. The work done is about **0.3 MB**, so the answer is option **(B)**.

← *Back to Q12*

Q13

Which of the following is a diamagnetic substance?

(A) Gadolinium

(B) Sodium

(C) Copper chloride

(D) **Sodium chloride**

[2024 • Set 55-3-3]

SOLUTION

A diamagnetic substance is one in which every electron is paired, so the atom or ion has no permanent magnetic dipole moment. It only develops a weak induced moment opposite to the applied field.

Checking each option against this rule helps fix the answer.

Gadolinium has several unpaired 4f electrons and is in fact ferromagnetic near room temperature, so option (A) is rejected.

Sodium has one unpaired electron in its 3s orbital, giving it a permanent moment, so it is paramagnetic and option (B) is rejected.

In copper chloride, the Cu^{2+} ion has an unpaired electron in its 3d subshell, so it carries a permanent moment and is paramagnetic; option (C) is rejected.

In sodium chloride, both Na^+ and Cl^- have completely filled shells with all electrons paired, so the moment is zero and the salt is diamagnetic.

Hence the diamagnetic substance is **sodium chloride**, which is option (D).

Final answer. Sodium chloride is the diamagnetic substance, so the answer is option **(D)**.

← *Back to Q13*

Q14

The magnetic susceptibility for a diamagnetic material is

(A) small and negative

(B) small and positive

(C) large and negative

(D) large and positive

[2024 • Set 55-4-1]

SOLUTION

Magnetic susceptibility χ_m links the magnetisation $\vec{M}$ of a material to the magnetising field $\vec{H}$ through $\vec{M} = \chi_m \vec{H}$. Its sign and size classify the material.

In a diamagnetic substance the applied field induces a moment that opposes the field (a consequence of Lenz's law acting on orbital electrons). Because the magnetisation points opposite to $\vec{H}$, the susceptibility is negative.

The induced response is also extremely feeble, so the magnitude is tiny, of the order of $\chi_m \approx -10^{-5}$.

Therefore the susceptibility of a diamagnetic material is **small and negative**, which is option (A).

For comparison, a paramagnetic material has small positive χ_m (about $+10^{-5}$ to $+10^{-3}$), and a ferromagnetic material has large positive χ_m (about 10^3 to 10^5). This rules out options (B), (C) and (D), since only diamagnets carry a negative susceptibility, and a small one at that.

Final answer. The susceptibility of a diamagnetic material is small and negative, so the answer is option **(A)**.

← *Back to Q14*

Q15

Which of the following pairs is that of paramagnetic materials?

(A) Copper and Aluminium

(B) Sodium and Calcium

(C) Lead and Iron

(D) Nickel and Cobalt

[2024 • Set 55-5-1]

SOLUTION

Paramagnetic materials have atoms with a permanent magnetic moment (at least one unpaired electron). They show a small positive susceptibility, of order $\chi_m \approx +10^{-5}$ to $+10^{-3}$, because the moments align partly with the applied field.

To answer, each material in the options must be classified.

Copper is diamagnetic and lead is also diamagnetic. Iron, nickel and cobalt are the classic ferromagnetic metals. Aluminium, sodium and calcium are paramagnetic.

Now testing the pairs. Option (A) Copper and Aluminium pairs a diamagnet with a paramagnet, so it fails. Option (C) Lead and Iron pairs a diamagnet with a ferromagnet, so it fails. Option (D) Nickel and Cobalt are both ferromagnetic, not paramagnetic, so it fails.

Only option (B), Sodium and Calcium, is a pair where both members are paramagnetic.

Hence the paramagnetic pair is **Sodium and Calcium**, option (B).

Final answer. Sodium and Calcium form the paramagnetic pair, so the answer is option **(B)**.

← Back to Q15

Q16

An iron needle is kept near a strong bar magnet. It will experience:

(A) a force of attraction and no torque.

(B) a force of attraction and a torque.

(C) a torque and no force.

(D) neither a force nor a torque.

[2024 • Set 55-5-2]

SOLUTION

An iron needle is ferromagnetic. When it is brought near a strong bar magnet, the non-uniform field of the bar magnet magnetises the needle by induction.

The end of the needle nearer the magnet develops a pole opposite to the nearby pole of the bar magnet. Since unlike poles attract and the field is stronger at the near end than the far end, the net result is a force of attraction pulling the needle toward the magnet.

The needle behaves like a small magnetic dipole of moment $\vec{m}$. If its axis is not already lined up with the local field $\vec{B}$, it also feels a torque $\vec{\tau} = \vec{m} \times \vec{B}$, of magnitude $\tau = mB\theta$, which tends to turn it into alignment with the field.

So the needle experiences both a force of attraction (because the field is non-uniform) and a torque (because of the angle between its moment and the field).

The correct choice is therefore **option (B): a force of attraction and a torque**.

Options (A) and (C) are incomplete, since each drops one of the two effects, and (D) is wrong because the strong nearby magnet clearly acts on the iron. Only (B) captures both effects.

Final answer. The needle experiences both a force of attraction and a torque, so the answer is option **(B)**.

← Back to Q16

Q17

Which of the following material has its magnetic susceptibility χ in the range $0 < \chi < \varepsilon$, where ε is positive and small?

(A) Aluminium

(B) Water

(C) Gadolinium

(D) Bismuth

[2024 • Set 55-5-3]

SOLUTION

The condition $0 < \chi < \varepsilon$ for a small positive number ε describes a material whose susceptibility is small and positive. This is the defining signature of a paramagnetic material.

Each option must be classified by the sign and size of its susceptibility.

Aluminium is paramagnetic, with $\chi \approx +2\times 10^{-5}$, which is small and positive, so it satisfies the condition.

Water is diamagnetic, with $\chi \approx -9\times 10^{-6}$, a negative value, so it is rejected.

Gadolinium is ferromagnetic, with $\chi \gg 1$, far too large, so it is rejected.

Bismuth is strongly diamagnetic, with $\chi \approx -1.6\times 10^{-4}$, again negative, so it is rejected.

Only aluminium has a small positive susceptibility lying in the required range.

Hence the material is **Aluminium**, option (A).

Final answer. Aluminium has a small positive susceptibility in the given range, so the answer is option **(A)**.

← *Back to Q17*

Q18

Assertion (A): *When a bar of copper is placed in an external magnetic field, the field lines get concentrated inside the bar.*
Reason (R): *Copper is a paramagnetic substance.*

(D) Assertion (A) is false and Reason (R) is also false.

[2023 • Set 55-1-1]

SOLUTION

This is an assertion-reason question, so the Assertion and the Reason must each be tested for truth.

Copper is a diamagnetic substance, not paramagnetic. When a diamagnetic material is placed in an external field, the magnetic field lines are pushed apart and partly expelled from the material, because its relative permeability is slightly less than one ($\mu_r < 1$).

The Assertion claims the field lines get concentrated inside the bar. That is what happens for a paramagnetic or ferromagnetic bar, not a diamagnetic one. So for copper the lines actually spread out, and the Assertion is false.

The Reason states that copper is paramagnetic. Copper is in fact diamagnetic, so the Reason is also false.

Since both statements are false, the correct choice is **option (D): Assertion is false and Reason is also false**.

Options (A), (B) and (C) all require at least one of the statements to be true, which is not the case here, so they are ruled out.

COMMON MISTAKE

Remember the field-line picture: diamagnets push lines out ($\mu_r<1$), paramagnets and ferromagnets pull lines in ($\mu_r>1$). Copper sits in the diamagnetic group with Bi, water and NaCl.

Final answer. Both the Assertion and the Reason are false, so the answer is option **(D)**.

← Back to Q18

Q19

A diamagnetic substance is brought near the north or south pole of a bar magnet. It will be:

(A) repelled by both the poles.

(B) attracted by both the poles.

(C) repelled by the north pole and attracted by the south pole.

(D) attracted by the north pole and repelled by the south pole.

[2023 • Set 55-1-1]

SOLUTION

A diamagnetic substance develops an induced magnetic moment that points opposite to the applied field, following Lenz's law applied to the orbital motion of its electrons.

Bring the substance near the north pole of the bar magnet. The field there points outward, away from the pole, so the induced moment points the other way. This makes the face of the substance nearest the magnet behave like a north pole. Two north poles face each other, and like poles repel, so the substance is pushed away.

Now bring the substance near the south pole. The field there points inward, toward the pole, so the induced moment again opposes it. This makes the near face behave like a south pole. Two south poles face each other, and again like poles repel.

In both cases the induced pole is the same kind as the nearby pole of the magnet, so the substance is repelled either way.

Hence a diamagnetic substance is **repelled by both the poles**, which is option (A).

This is a defining test for diamagnetism. The other options describe attraction or a mixed response, which belong to para or ferromagnetic behaviour, so (B), (C) and (D) are rejected.

Final answer. A diamagnetic substance is repelled by both poles, so the answer is option **(A)**.

← *Back to Q19*

Q20

Assertion (A): *Diamagnetic substances exhibit magnetism.*
Reason (R): *Diamagnetic materials do not have permanent magnetic dipole moment.*

(B) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

[2023 • Set 55-2-1]

SOLUTION

This assertion-reason question needs each statement judged separately, and then the link between them tested.

The Assertion says diamagnetic substances exhibit magnetism. They do show a weak induced magnetism: in an applied field they develop a small magnetisation directed opposite to the field. So the Assertion is true.

The Reason says diamagnetic materials do not have a permanent magnetic dipole moment. This is correct, because in a diamagnetic atom all electrons are paired and their moments cancel, leaving no net permanent moment. So the Reason is true.

Now the key step: does the Reason explain the Assertion? The Reason only tells us why diamagnets have no spontaneous or permanent magnetism. It does not tell us why they show induced magnetism. The induced diamagnetic response comes from the change in the orbital motion of the electrons when the field is switched on (Lenz's law at the atomic level), a different cause altogether.

So both statements are true, but the Reason is not the correct explanation of the Assertion.

The correct choice is therefore **option (B)**.

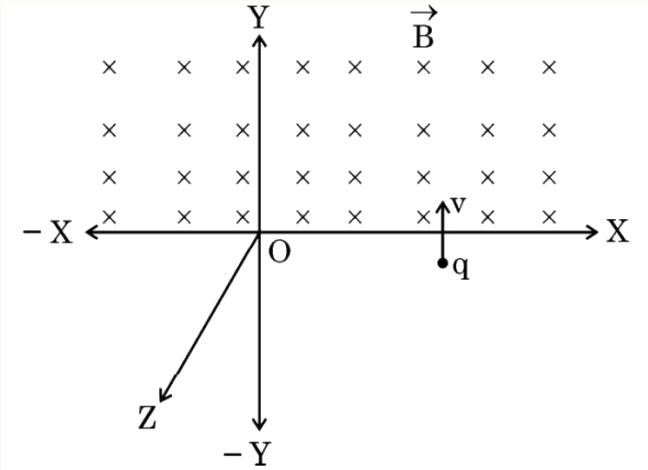
Option (A) is wrong because the explanation does not match, and options (C) and (D) are wrong because they call one or both statements false.

Final answer. Both A and R are true but R is not the correct explanation of A, so the answer is option **(B)**.

← *Back to Q20*

Q21

The magnetic field lines near a substance are as shown in the figure. The substance is:


Y
B
X
X
X

and crowd them. Sodium is paramagnetic, so it would weakly draw the lines in. Aluminium is also paramagnetic, again drawing the lines in slightly.

Only the diamagnetic substance expels the lines, so the material is Copper, option (A).

Final answer. The substance is Copper, option **(A)**.

← *Back to Q21*

Q22

Which one of the following has negative value of susceptibility?

(A) Nickel

(B) Aluminium

(C) **Lead**

(D) Iron

[2023 • Set 55-4-2]

SOLUTION

Magnetic susceptibility χ_m tells us how a material responds to an applied field. For diamagnetic materials χ_m is small and negative, for paramagnetic materials it is small and positive, and for ferromagnetic materials it is large and positive.

So the material with a negative susceptibility must be diamagnetic.

Classifying the options: Nickel is ferromagnetic with $\chi_m \gg 1$ (positive). Aluminium is paramagnetic with $\boldsymbol{\chi_m > 0}$. Lead is diamagnetic with $\boldsymbol{\chi_m \approx -1.6 \times 10^{-5}}$, which is negative. Iron is ferromagnetic with $\chi_m \gg 1$ (positive).

The only substance with a negative value of susceptibility is Lead, option (C).

Final answer. Lead has a negative susceptibility, option **(C)**.

← *Back to Q22*

Q23

Which one of the following has relative magnetic permeability between 0 and 1?

(A) Aluminium

(B) Alnico

(C) **Water**

(D) Sodium

[2023 • Set 55-4-3]

SOLUTION

Relative magnetic permeability is related to susceptibility by $\mu_r = 1 + \chi_m$.

For a diamagnetic material χ_m is small and negative, so μ_r is slightly less than 1, giving $0 < \mu_r < 1$.

For paramagnetic materials $\chi_m > 0$ so $\mu_r > 1$, and for ferromagnetic materials $\mu_r \gg 1$.

Checking the options: Aluminium is paramagnetic with μ_r slightly greater than 1. Alnico is ferromagnetic with $\mu_r \gg 1$. Water is diamagnetic with $\mu_r \approx 0.99999$, which lies between 0 and 1. Sodium is paramagnetic with $\mu_r > 1$.

Only the diamagnetic substance has $0 < \mu_r < 1$, so the answer is Water, option (C).

TIP

Remember the simple chain: diamagnetic gives $\chi_m<0$, hence $\mu_r<1$. Any "μ_r between 0 and 1" question is really asking which option is diamagnetic.

Final answer. Water has relative permeability between 0 and 1, option **(C)**.

← *Back to Q23*

Q24

Which of the following has its permeability less than that of free space?

(A) Copper

(B) Aluminium

(C) Copper chloride

(D) Nickel

[2023 • Set 55-5-1]

SOLUTION

The permeability of a material is $\mu = \mu_r \mu_0$, where μ_0 is the permeability of free space.

A material whose permeability is less than that of free space must have $\mu_r < 1$, which means $\chi_m < 0$. This is the property of a diamagnetic substance.

Checking the options: Copper is diamagnetic with $\mu_r < 1$. Aluminium is paramagnetic with $\mu_r > 1$. Copper chloride is paramagnetic, because the Cu^{2+} ion has an unpaired d-electron, so $\mu_r > 1$. Nickel is ferromagnetic with $\mu_r \gg 1$.

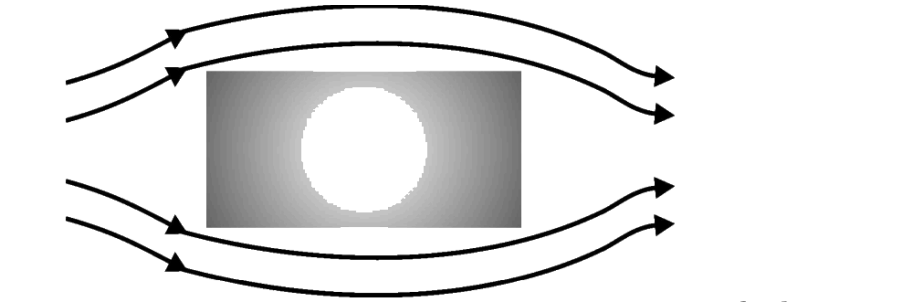
Only the diamagnetic substance has $\mu < \mu_0$, so the answer is Copper, option (A).

Final answer. Copper has permeability less than free space, option **(A)**.

← *Back to Q24*

Q25

Which of the following cannot modify an external magnetic field as shown in the figure?



(A) Nickel

(B) Silicon

(C) Sodium Chloride

(D) Copper

[2023 • Set 55-5-3]

SOLUTION

When a substance is placed in an external magnetic field, the field lines get modified according to the material's susceptibility χ_m. A ferromagnetic material strongly concentrates the lines, while paramagnetic and diamagnetic materials change them only slightly. A substance that effectively cannot modify the field is one whose magnetic response is so small that it behaves as nearly non-magnetic.

Checking the options: Nickel is strongly ferromagnetic with $\chi_m \gg 1$, so it modifies the field lines a great deal. Silicon has an extremely small diamagnetic susceptibility, $\chi_m \approx -4 \times 10^{-6}$, so for practical purposes it leaves the field essentially unchanged. Sodium chloride is diamagnetic and modifies the field slightly. Copper is diamagnetic and also modifies the field slightly.

Of these, silicon has by far the smallest magnetic effect and so does not appreciably modify the external field. The answer is Silicon, option (B).

Final answer. Silicon cannot modify the external field appreciably, option **(B)**.

← *Back to Q25*

Q26

Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as

given below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false and Reason (R) is also false.
Assertion (A): *In most of the northern hemisphere, the north pole of the dip needle tilts upwards.*
Reason (R): *This gives the direction of magnetic field of the Earth at that place.*

[2021]

SOLUTION

This is an Assertion-Reason question, so test each statement on its own first.

Test the Assertion. In most of the northern hemisphere the Earth's magnetic field points downward into the ground (the field lines dip into the Earth as we move north). A freely suspended dip needle aligns along this field, so its north pole tilts downward, not upward. The Assertion claims the north pole tilts upward, so the Assertion is false.

Test the Reason. A freely suspended magnetic needle always sets itself along the direction of the local magnetic field. So the orientation of the dip needle does correctly indicate the direction of the Earth's magnetic field at that place. Hence the Reason is true.

So we have a false Assertion with a true Reason. None of the four printed codes (A), (B), (C), (D) describes the pairing false-A with true-R exactly: codes (A) and (B) need both true, code (C) needs A true, and code (D) needs both false. So strictly no single code fits; the correct physics conclusion is that Assertion (*A*) is false while Reason (*R*) is true.

COMMON MISTAKE

Common mistake: students confuse geographic and magnetic poles. In the northern hemisphere the field dips into the ground, so the needle's north end goes down, not up.

Final answer. Assertion **(A)** is false (the dip needle's north pole tilts downward, not upward, in the northern hemisphere) and Reason (R) is true. None of the

four codes matches the false-A, true-R pairing exactly.

← Back to Q26

Q27

The magnetic field and angle of dip at a place on the earth are 0·3 *G and* 30°, *respectively. The value of vertical component of the earth's magnetic field at the place is* _____.

[2020 • Set 55-1-1]

SOLUTION

The Earth's total magnetic field B makes an angle of dip δ with the horizontal. The vertical component of the field is given by the relation $\mathbf{B_V = B\,\delta}$.

Here $\boldsymbol{B = 0.3\ G}$ and $\boldsymbol{\delta = 30°}$.

$$B_V = B\,\delta$$

$$B_V = 0.3 \times \sin 30°$$

$$B_V = 0.3 \times 0.5$$

$$B_V = 0.15\ G$$

Converting to SI units, since $\mathbf{1\ G = 10^{-4}\ T}$, we get $\mathbf{B_V = 0.15 \times 10^{-4}\ T = 1.5 \times 10^{-5}\ T}$.

The vertical component is $B_V = 0.15\ G = 1.5 \times 10^{-5}\ T$.

Final answer. The vertical component is $\mathbf{B_V = 0.15\ G = 1.510^{-5}\ T}$.

← Back to Q27

Q28

The magnetic field lines are _____ *by a diamagnetic substance.*

[2020 • Set 55-1-2]

SOLUTION

A diamagnetic substance has a small negative susceptibility, $\chi_m < 0$. When placed in an external field it sets up a weak magnetisation in the opposite direction to the applied field.

Because of this opposing magnetisation, the field inside the substance becomes weaker than outside, and the magnetic field lines are pushed out of the interior. They bend around the sample instead of passing through it.

Hence the magnetic field lines are repelled (expelled) by a diamagnetic substance.

Final answer. The magnetic field lines are repelled (expelled) by a diamagnetic substance.

← *Back to Q28*

Q29

The material which is not suitable for making a permanent magnet is

(A) Steel

(B) Ticonal

(C) Lead

(D) Alnico

[2020 • Set 55-2-2]

SOLUTION

A material suitable for making a permanent magnet must satisfy three conditions: it must be ferromagnetic, it must have high retentivity so that it stays strongly magnetised after the external field is removed, and it must have high coercivity so that it resists demagnetisation.

Checking the options: Steel is ferromagnetic with high retentivity and high coercivity, so it is suitable. Ticonal is a ferromagnetic alloy, so it is suitable. Lead is diamagnetic with $\chi_m < 0$, so it cannot be magnetised at all and is not suitable. Alnico is a ferromagnetic alloy used for strong permanent magnets, so it is suitable.

The only material that cannot be used for a permanent magnet is the diamagnetic one, namely Lead, option (C).

Final answer. Lead is not suitable for a permanent magnet, option **(C)**.

← *Back to Q29*

Q30

Define 'magnetic declination' at a place on earth.

[2020 • Set 55-4-1]

SOLUTION

At any place on the Earth two vertical planes can be drawn. The geographic meridian is the vertical plane that contains the geographic north-south axis (the direction of true north). The magnetic meridian is the vertical plane that contains the Earth's magnetic field at that place, that is, the direction shown by a freely suspended compass needle.

These two planes generally do not coincide.

The angle between the geographic meridian and the magnetic meridian at a place is called the magnetic declination at that place.

Equivalently, it is the angle between the direction of true (geographic) north and the direction of magnetic north. Its value differs from one place to another and also changes slowly with time as the Earth's magnetic field drifts.

Final answer. Magnetic declination is the angle between the geographic meridian and the magnetic meridian at a place.

← *Back to Q30*

Q31

Define 'angle of dip' at a place on earth.

[2020 • Set 55-4-2]

SOLUTION

The angle of dip (also called magnetic inclination) at a place is defined as the angle that the direction of the Earth's total magnetic field $\vec{B}$ makes with the horizontal direction at that place. It is measured in the magnetic meridian, that is, in the vertical plane that contains the Earth's magnetic field. It is denoted by δ.

If B_H is the horizontal component and B_V is the vertical component of the Earth's magnetic field, then

$$\delta = \frac{B_V}{B_H}$$

At the magnetic equator the field is horizontal, so $\delta = 0°$, while at the magnetic poles the field is vertical, so $\delta = 90°$. Thus the **angle of dip is the angle between the Earth's total magnetic field and the horizontal at a given place.**

Final answer. The angle of dip δ is the angle that the Earth's total magnetic field makes with the horizontal in the magnetic meridian at a place.

← *Back to Q31*

Q32

The magnetic susceptibility of magnesium at 300 *K is* 1.2×10^{-5}. *At what temperature will its magnetic susceptibility become* 1.44×10^{-5}*?*

OR

The magnetic susceptibility χ *of a given material is* -0.5. *Identify the magnetic material.*

[2019 • Set 55-2-1]

SOLUTION

Magnesium is a paramagnetic material, so its susceptibility follows Curie's law,

$$\chi = \frac{C}{T}$$

where C is the Curie constant. Hence the product χT stays constant, giving

$$\chi_1 T_1 = \chi_2 T_2$$

Here $\chi_1 = 1.2\times 10^{-5}$ at $T_1 = 300\ K$, and $\chi_2 = 1.44\times 10^{-5}$. Solving for T_2,

$$T_2 = T_1 \times \frac{\chi_1}{\chi_2}$$

$$T_2 = 300\times \frac{1.2\times 10^{-5}}{1.44\times 10^{-5}}\ K$$

$$T_2 = 300\times \frac{1.2}{1.44}\ K = \mathbf{250\ K}$$

So the susceptibility of magnesium becomes 1.44×10^{-5} at 250 K.

OR

For the material with susceptibility $\chi = -0.5$: the susceptibility is negative, which is the characteristic sign of a **diamagnetic material**. A negative χ means the induced magnetisation opposes the applied field, so the value $\chi = -0.5$ identifies a (strongly) diamagnetic substance.

CONCEPT

Curie's law applies only to paramagnetic materials, so use χT = constant only when the substance (here magnesium) is paramagnetic, never for diamagnets.

Final answer. $T_2 = 250\ K$; OR a material with $\chi = -0.5$ is diamagnetic.

← *Back to Q32*

Q33

Write one important property of a paramagnetic material.
OR
Do the diamagnetic substances have resultant magnetic moment in an atom in the absence of external magnetic field?

[2019 • Set 55-5-1]

SOLUTION

One important property of a paramagnetic material is that it has a small, positive magnetic susceptibility, χ_m of the order of $+10^{-3}$ to $+10^{-5}$. Because of this, when a paramagnetic material is placed in an external magnetic field it is weakly attracted towards the region of stronger field, and the field lines become slightly concentrated inside it. Its magnetisation $\vec{M}$ is in the same direction as the applied field $\vec{B}$.

(Other acceptable properties: relative permeability μ_r slightly greater than 1; susceptibility obeys Curie's law $\chi \propto \frac{1}{T}$; its atoms carry permanent magnetic dipole moments.)

OR

Do diamagnetic substances have a resultant magnetic moment per atom in the absence of an external field?

No. In a diamagnetic substance all the electrons are paired, so the orbital and spin magnetic moments of the electrons cancel completely. Therefore, in the absence of an external magnetic field, each atom has **zero net magnetic moment.** A magnetic moment opposing the field is induced only when an external field is applied.

Final answer. A paramagnetic material has small positive susceptibility and is weakly attracted by a magnetic field; OR diamagnetic atoms have zero resultant magnetic moment in the absence of an external field.

← *Back to Q33*

Q34

At a place, the horizontal component of earth's magnetic field is B and angle of dip is 60°. What is the value of horizontal component of the earth's magnetic field at equator?

[2017]

SOLUTION

The horizontal component of the Earth's field is related to the total field by

$$B_H = B_{\text{total}}\delta$$

At the given place, $\delta = 60°$ and $B_H = B$. So the total field there is

$$B_{\text{total}} = \frac{B_H}{\delta} = \frac{B}{\cos 60°}$$

$$B_{\text{total}} = \frac{B}{0.5} = 2B$$

Taking the magnitude of the Earth's total field to be the same up to the equator, at the magnetic equator the angle of dip is $\delta = 0°$, so the field is entirely horizontal. Hence the horizontal component at the equator is

$$B_H^{(\text{eq})} = B_{\text{total}}\cos 0° = B_{\text{total}} \times 1 = 2B$$

So the horizontal component of the Earth's magnetic field at the equator is **2B**.

Final answer. The horizontal component of the Earth's magnetic field at the equator is **2B**.

← *Back to Q34*

Q35

In what way is the behaviour of a diamagnetic material different from that of a paramagnetic, when kept in an external magnetic field?

[2016]

SOLUTION

When kept in an external magnetic field, the two materials behave oppositely. The key differences are:

In a diamagnetic material the susceptibility χ_m is small and negative, the magnetisation $\vec{M}$ is set up opposite to the applied field $\vec{B}$, the field lines are pushed out (expelled) from the material, and in a non-uniform field it is weakly repelled, moving towards the weaker field region.

In a paramagnetic material the susceptibility χ_m is small and positive, the magnetisation $\vec{M}$ is in the same direction as $\vec{B}$, the field lines are slightly drawn

into (concentrated within) the material, and in a non-uniform field it is weakly attracted, moving towards the stronger field region.

The reason is that diamagnetic atoms have no permanent magnetic moment, and an applied field induces a moment opposing itself; paramagnetic atoms have permanent moments that partially align with the field. Thus **a diamagnetic material is weakly repelled while a paramagnetic material is weakly attracted by an external magnetic field.**

Final answer. A diamagnetic material is repelled and magnetised opposite to the field, whereas a paramagnetic material is attracted and magnetised along the field.

← *Back to Q35*

Q36

What are permanent magnets? Give one example.

[2013]

SOLUTION

Permanent magnets are materials that retain their magnetisation even after the external magnetising field is removed. They are made from ferromagnetic substances having high retentivity (so they stay strongly magnetised) and high coercivity (so they are not easily demagnetised). Such materials have a broad hysteresis loop.

Once magnetised, a permanent magnet keeps producing a magnetic field around itself without needing any external current or field.

Example: steel (other valid examples are Alnico, an alloy of aluminium, nickel, cobalt and iron, or a neodymium magnet).

Thus a **permanent magnet retains its magnetism after the magnetising field is removed, for example steel.**

Final answer. Permanent magnets retain their magnetisation after the magnetising field is removed; example: steel.

← Back to Q36

Q37

The horizontal component of the earth's magnetic field at a place is B and angle of dip is 60°. What is the value of vertical component of earth's magnetic field at equator?

[2012]

SOLUTION

At a place, the vertical and horizontal components of the Earth's field are related to the dip by

$$\delta = \frac{B_V}{B_H}$$

Given $\delta = 60°$ and $B_H = B$, the total field magnitude is

$$B_{\text{total}} = \frac{B_H}{\delta} = \frac{B}{\cos 60° = \frac{B}{0.5} = 2B}$$

At the magnetic equator the angle of dip is $\delta = 0°$, so the Earth's field is purely horizontal. The vertical component there is

$$B_V^{(eq)} = B_{\text{total}} \sin 0° = 2B \times 0$$

$$B_V^{(eq)} = 0$$

So the vertical component of the Earth's magnetic field at the equator is **zero.**

Final answer. The vertical component of the Earth's magnetic field at the equator is zero.

← Back to Q37

Q38

A magnetic needle, free to rotate in a vertical plane, orients itself vertically at a certain place on the Earth. What are the values of (i) horizontal component of Earth's magnetic field and (ii) angle of dip at this place?

[2012]

SOLUTION

A freely suspended magnetic needle always sets itself along the direction of the local magnetic field. Here the needle, free to rotate in a vertical plane, orients itself vertically, which means the Earth's magnetic field at that place is directed entirely vertically (it has no horizontal part).

(i) Since the field has no horizontal part, the horizontal component is

$$B_H = 0$$

(ii) The angle of dip is the angle of the field with the horizontal. As the field is vertical,

$$\delta = 90^\circ$$

So at this place $\mathbf{B_H = 0}$ and $\boldsymbol{\delta = 90^\circ}$, which corresponds to one of the Earth's magnetic poles.

Final answer. At this place the horizontal component $\mathbf{B_H = 0}$ and the angle of dip $\boldsymbol{\delta = 90^\circ}$.

← *Back to Q38*

Q39

Where on the surface of Earth is the angle of dip 90°*?*

[2011]

SOLUTION

The angle of dip δ is the angle that the Earth's magnetic field makes with the horizontal. The dip is 90° only where the Earth's magnetic field is directed vertically, that is, perpendicular to the surface.

This happens at the **magnetic poles** of the Earth, where the field lines enter or leave the surface vertically. (The magnetic poles lie near, but not exactly at, the geographic poles.)

Therefore the angle of dip is 90° at the Earth's magnetic poles.

Final answer. The angle of dip is **90°** at the Earth's magnetic poles.

← Back to Q39

Q40

The permeability of a magnetic material is 0.9983. *Name the type of magnetic materials it represents.*

[2011 • Set 55-1-1]

SOLUTION

The relative permeability of the material is found from its permeability. Here the given (relative) permeability is

$$\mu_r = 0.9983$$

The magnetic susceptibility is related to relative permeability by

$$\chi_m = \mu_r - 1$$

$$\chi_m = 0.9983 - 1 = -0.0017$$

Since μ_r is slightly less than 1 and the susceptibility χ_m is small and negative, the material is **diamagnetic.**

(Examples of diamagnetic materials: copper, bismuth, water, sodium chloride.)

TIP

A relative permeability just below 1 (so χ_m small and negative) means diamagnetic; just above 1 means paramagnetic; much greater than 1 means ferromagnetic.

Final answer. The material is diamagnetic, since $\mu_r < 1$ and $\chi_m = -0.0017$ is small and negative.

← *Back to Q40*

Q41

Why should the material used for making permanent magnets have high coercivity?

[2007]

SOLUTION

Coercivity is the value of the reverse (demagnetising) magnetic field needed to reduce the magnetisation of a fully magnetised material to zero. A material with high coercivity therefore strongly resists being demagnetised.

A permanent magnet must keep its magnetism for a long time, even when it is exposed to external influences. If the coercivity were low, stray magnetic fields from nearby magnets or current-carrying wires, mechanical shocks, vibrations, and temperature changes would easily wipe out its magnetisation.

Hence the material used for a permanent magnet must have **high coercivity** so that the magnet retains its magnetisation throughout its working life and is not demagnetised by ordinary external disturbances.

Final answer. A high coercivity makes the magnet strongly resist demagnetisation, so it keeps its magnetism permanently.

← *Back to Q41*

Q42

The vertical component of Earth's magnetic field at a place is $\sqrt{3}$ times the horizontal component. What is the value of angle of dip at this place?

[2006]

SOLUTION

The angle of dip δ at a place is the angle that the Earth's total magnetic field makes with the horizontal. It is related to the vertical component B_V and the horizontal component B_H by:

$$\delta = \frac{B_V}{B_H}$$

Here it is given that $\mathbf{B_V = \sqrt{3}\ B_H}$.

Substituting,

$$\delta = \frac{\sqrt{3}\ B_H}{B_H} = \sqrt{3}$$

$$\Rightarrow \delta = \tan^{-1}(\sqrt{3})$$

$$\Rightarrow \delta = 60^\circ$$

Hence the angle of dip at this place is **60°**.

Final answer. The angle of dip at the place is **60°**.

← *Back to Q42*

Q43

How does the (i) pole strength and (ii) magnetic moment of each part of a bar magnet change if it is cut into two equal pieces transverse to its length?

[2003]

SOLUTION

Let the original bar magnet have pole strength q_m and magnetic length 2, so its magnetic moment is:

$$m = q_m \times 2$$

When the magnet is cut transverse (perpendicular) to its length into two equal pieces, each piece has the same cross-sectional area but half the original length, that is a magnetic length of .

(i) Pole strength: The pole strength depends on the cross-sectional area of the magnet (the magnetic charge spread over the end face). Since cutting transverse to the length does not change the cross-section, the pole strength of each piece stays the same.

$\mathbf{q_m' = q_m}$ (unchanged)

(ii) Magnetic moment of each piece: Each piece now has pole strength q_m but magnetic length , so:

$$m' = q_m \times = \frac{q_m \times 2}{2} = \frac{m}{2}$$

Hence the **pole strength of each piece remains q_m (unchanged)**, while the **magnetic moment of each piece becomes $\frac{m}{2}$** (halved).

COMMON MISTAKE

Cutting transverse to the length only halves the length, not the cross-section, so pole strength stays the same. A cut along the length would halve the pole strength instead.

Final answer. Pole strength of each piece is unchanged; magnetic moment of each piece becomes half ($m/2$).

← *Back to Q43*

Q44

A small magnetised needle P is placed at the origin of x-y plane with its magnetic moment pointing along the y-axis. Another identical magnetised needle Q is placed in two positions, one by one.

Case 1: at $(a, 0)$ *with its magnetic moment pointing along x-axis.*
Case 2: at $(0, a)$ *with its magnetic moment pointing along y-axis.*
(a) In which case is the potential energy of P and Q minimum?
(b) In which case is P and Q not in equilibrium?
Justify your answers.

[2023 • Set 55-4-1]

SOLUTION

Needle P is at the origin with magnetic moment $\vec{m}_P = M\hat{j}$ (along the y-axis). The identical needle Q (moment magnitude M) is placed in two cases. The magnetic field of a dipole $\vec{m}$ at distance r is:

on its axis, $\vec{B} = \frac{\mu_0}{4\pi}\frac{2\vec{m}}{r^3}$ (along $\vec{m}$),

on its equatorial line, $\vec{B} = -\frac{\mu_0}{4\pi}\frac{\vec{m}}{r^3}$ (opposite to $\vec{m}$).

The potential energy of Q in the field of P is $U = -\vec{m}_Q \cdot \vec{B}_P$.

Case 1: Q at $(a,0)$ with $\vec{m}_Q = M\hat{i}$. Since $\vec{m}_P$ is along $\hat{j}$, the point $(a,0)$ lies on the equatorial line of P, so the field of P there is:

$$\vec{B}_1 = -\frac{\mu_0}{4\pi}\frac{M}{a^3}\hat{j}$$

$$U_1 = -\vec{m}_Q \cdot \vec{B}_1 = -(M\hat{i}) \cdot \left(-\frac{\mu_0 M}{4\pi a^3}\hat{j}\right) = 0$$

Case 2: Q at $(0,a)$ with $\vec{m}_Q = M\hat{j}$. The point $(0,a)$ lies on the axis of P, so the field of P there is:

$$\vec{B}_2 = \frac{\mu_0}{4\pi}\frac{2M}{a^3}\hat{j}$$

$$U_2 = -\vec{m}_Q \cdot \vec{B}_2 = -(M\hat{j}) \cdot \frac{\mu_0 (2M)}{4\pi a^3}\hat{j} = -\frac{\mu_0 M^2}{2\pi a^3}$$

(a) Since U_2 is negative and $U_1 = 0$, the potential energy is minimum in **Case 2**.

(b) For equilibrium the torque $\vec{\tau} = \vec{m}_Q \times \vec{B}_P$ must be zero, which needs $\vec{m}_Q$ parallel (or anti-parallel) to $\vec{B}_P$. In Case 1, $\vec{m}_Q = M\hat{i}$ but $\vec{B}_1$ is along $-\hat{j}$, so they are perpendicular and a torque acts on Q. Hence P and Q are **not in equilibrium in Case 1**. In Case 2, $\vec{m}_Q$ and $\vec{B}_2$ are both along $\hat{j}$ (parallel), so the torque is zero and Q is in stable equilibrium.

Final answer. Potential energy is minimum in Case 2; P and Q are not in equilibrium in Case 1.

← Back to Q44

Q45

Answer the following giving reasons:
(a) Do the magnetic field lines also represent the 'lines of force' on a moving charged particle at every point?
(b) If magnetic monopoles existed, how would the Gauss's law of magnetism be modified?

[2023 • Set 55-4-2]

SOLUTION

(a) No, the magnetic field lines do not represent the lines of force on a moving charged particle. The force on a charge q moving with velocity $\vec{v}$ in a magnetic field is:

$$\vec{F}=q\,\vec{v}\times\vec{B}$$

This force is always perpendicular to both $\vec{v}$ and $\vec{B}$. Since a field line gives the direction of $\vec{B}$, the force on the moving charge is not along the field line. (This is unlike the electric field, where $\vec{F}=q\vec{E}$ acts along the field line.)

(b) At present, since no isolated magnetic monopoles exist, Gauss's law for magnetism states that the net magnetic flux through any closed surface is zero:

$$\oint \vec{B}\cdot d\vec{S}=0$$

If magnetic monopoles existed, carrying a magnetic charge q_m, the law would be modified to be analogous to Gauss's law in electrostatics:

$$\oint \vec{B}\cdot d\vec{S}=\mu_0\, q_{m(\text{enclosed})}$$

where $q_{m(\text{enclosed})}$ is the total magnetic charge enclosed by the surface. Thus the right-hand side would no longer be zero whenever a net magnetic charge is enclosed.

Final answer. (a) No, since $\vec{F}=q\vec{v}\times\vec{B}$ is perpendicular to $\vec{B}$; (b) it becomes $\oint \vec{B}\cdot d\vec{S} = \mu_0 q_{m(\text{enclosed})}$.

← Back to Q45

Q46

(i) The angle of dip at a location in southern India is about 18°. *Would you expect a greater or smaller dip angle in Britain? Justify your answer.*
(ii) "The declination in India is small, it being 0°41' *E at Delhi and* 0°58' *W at Mumbai." What is the significance of this statement?*

[2021]

SOLUTION

(i) Angle of dip in Britain compared to southern India: The angle of dip is 0° at the magnetic equator and increases to 90° at the magnetic poles. Southern India lies close to the magnetic equator, which is why the dip there is small (about 18°). Britain is at a much higher magnetic latitude, far nearer to the north magnetic pole.

Therefore the angle of dip in Britain is much greater than in southern India (it is roughly 65° to 70° there).

(ii) Significance of the small declination in India: Magnetic declination at a place is the angle between the geographic meridian (true north-south line) and the magnetic meridian (direction shown by a freely suspended magnetic needle).

The quoted values are very small (0°41' E at Delhi and 0°58' W at Mumbai). This means that across India the magnetic north shown by a compass almost coincides with the true geographic north.

So a compass needle in India points very nearly towards true north, and it can be used for navigation with almost no correction for declination, unlike in many other parts of the world where the declination is large.

Final answer. (i) The dip angle in Britain is greater than in southern India; (ii) it means a compass in India points almost exactly to true (geographic) north.

← Back to Q46

Q47

The magnetic moment of a circular coil carrying current I, having N turns, each of radius r, is M. Find the magnetic moment of the same coil if it is unwound and rewound into a coil having 2N *turns for the same current.*

[2021]

SOLUTION

For a circular coil of N turns, each of radius r, carrying current I, the magnetic moment is:

$$M=N\,I\,A=N\,I\,(\pi\,r^2)$$

The total length of wire used in the coil is:

$$L=N\times(2\pi\,r)$$

When this same wire is unwound and rewound into a coil of 2N turns, the wire length stays the same. Let the new radius be r'. Then:

$$L=(2N)(2\pi\,r')$$

Equating the two expressions for L,

$$N(2\pi\,r)=(2N)(2\pi\,r')$$

$$\Rightarrow r'=\frac{N(2\pi\,r)}{2N(2\pi)}=\frac{r}{2}$$

The new magnetic moment for the same current I is:

$$M'=(2N)\,I\,\pi\,r'^2=(2N)\,I\,\pi(\frac{r}{2})^2$$

$$M'=(2N)\,I\,\pi\frac{r^2}{4}=\frac{N\,I\,\pi\,r^2}{2}$$

Since $\boldsymbol{M=N\,I\,\pi\,r^2}$,

$$M' = \frac{M}{2}$$

Hence the magnetic moment of the rewound coil is **half** of the original, that is $\frac{M}{2}$.

TIP

Keep the wire length fixed when a coil is rewound. More turns force a smaller radius, and since area depends on r^2, the moment drops even though the turns doubled.

Final answer. The magnetic moment of the rewound **2N**-turn coil is $M/2$.

← *Back to Q47*

Q48

Two identical bars, one of paramagnetic material and other of diamagnetic material are kept in a uniform external magnetic field parallel to it. Draw diagrammatically the modifications in the magnetic field pattern in each case.

[2020 • Set 55-3-1]

SOLUTION

When a bar is placed in a uniform external magnetic field $\vec{B}$ (with the field parallel to the bar's length), the field pattern changes according to the magnetic nature of the bar.

(i) Paramagnetic bar: A paramagnetic material has relative permeability slightly greater than one ($\mu_r > 1$), so it slightly concentrates the field. The field lines are drawn into the bar, and the field lines inside become slightly more crowded (denser) than the surrounding field.

Diagram: Draw uniform, equally spaced, parallel horizontal field lines coming from the left and going to the right. As they reach the rectangular paramagnetic bar (drawn horizontally, parallel to the field), the lines bend slightly inward towards the bar and pass through it, so that inside the bar the lines are packed a little closer together than outside.

(ii) Diamagnetic bar: A diamagnetic material has relative permeability slightly less than one ($\mu_r<1$), so it slightly repels the field. The field lines are pushed out of the bar, and the field inside becomes slightly weaker (rarer) than the surrounding field.

Diagram: Again draw uniform parallel horizontal field lines. As they meet the rectangular diamagnetic bar, the lines bend outward (away from the bar) and curve around it, so that inside the bar the lines are slightly farther apart (fewer lines) than outside.

This opposite behaviour follows from the signs of susceptibility: paramagnetic ($\chi>0$) pulls the lines in, while diamagnetic ($\chi<0$) pushes the lines out.

Final answer. Paramagnetic bar: field lines crowd into and through it; diamagnetic bar: field lines are pushed out and curve around it.

← *Back to Q48*

Q49

Write two properties of a material suitable for making (a) a permanent magnet, and (b) an electromagnet.

[2017]

SOLUTION

(a) Material for a permanent magnet should have:

1. High retentivity, so that it retains a strong magnetisation even after the magnetising (external) field is removed.

2. High coercivity, so that a large reverse field is needed to demagnetise it, and it does not lose its magnetism easily due to stray fields or shocks.

(Such materials, like steel, alnico and ferrites, have a broad hysteresis loop. A large hysteresis-loop area is also acceptable for such permanent-magnet use.)

(b) Material for an electromagnet should have:

1. High permeability, so that it produces a strong magnetic field even for a small magnetising current.

2. Low coercivity (low retentivity), so that it loses almost all its magnetism the moment the current is switched off, and does not stay magnetised.

(Soft iron is the common choice, as it has a narrow hysteresis loop satisfying both needs.)

Final answer. Permanent magnet: high retentivity and high coercivity; electromagnet: high permeability and low coercivity (low retentivity).

← Back to Q49

Q50

Out of the two magnetic materials, 'A' has relative permeability slightly greater than unity while 'B' has less than unity. Identify the nature of the materials 'A' and 'B'. Will their susceptibilities be positive or negative ?

[2014]

SOLUTION

The relative permeability μ_r and the magnetic susceptibility χ_m of a material are related by:

$$\mu_r = 1 + \chi_m$$

Material A: Its relative permeability is slightly greater than unity, $\mu_r > 1$. Therefore:

$\chi_m = \mu_r - 1 > 0$ (small and positive)

A material with a small positive susceptibility is a paramagnetic material. So A is **paramagnetic**, with a positive susceptibility.

Material B: Its relative permeability is slightly less than unity, $\mu_r < 1$. Therefore:

$\chi_m = \mu_r - 1 < 0$ (small and negative)

A material with a small negative susceptibility is a diamagnetic material. So B is **diamagnetic**, with a negative susceptibility.

Hence A is paramagnetic (susceptibility positive) and B is diamagnetic (susceptibility negative).

Final answer. A is paramagnetic with positive susceptibility; B is diamagnetic with negative susceptibility.

← *Back to Q50*

Q51

Show diagrammatically the behaviour of magnetic field lines in the presence of (i) paramagnetic and (ii) diamagnetic substances. How does one explain this distinguishing feature ?

[2014]

SOLUTION

Diagrammatic behaviour of the field lines:

(i) Paramagnetic substance: Draw a uniform horizontal magnetic field with evenly spaced straight field lines pointing left to right. Place an oval-shaped paramagnetic sample in the middle. The lines bend slightly inward and crowd together inside the sample, so the line density inside is a little greater than outside. The sample is feebly attracted into the field.

(ii) Diamagnetic substance: Draw the same uniform field. Place an oval-shaped diamagnetic sample in the middle. Here the lines bow outward and avoid the sample, so the line density inside is less than outside. Some lines are expelled and bend around the body. The sample is feebly repelled by the field.

Explanation of the distinguishing feature:

In a paramagnetic substance the atoms possess permanent magnetic dipole moments. The external field $\vec{B}$ partially aligns these atomic dipoles parallel to itself, producing a magnetisation $\vec{M}$ in the direction of $\vec{B}$. Hence the field inside is enhanced and the lines are drawn in.

In a diamagnetic substance the atoms have no net magnetic moment (all electron spins paired). When $\vec{B}$ is applied, the orbital electron motion is altered and, by Lenz's law at the atomic level, an induced moment opposite to $\vec{B}$ appears. This gives a magnetisation $\vec{M}$ antiparallel to $\vec{B}$, so the field inside is reduced and the lines are pushed out.

The distinguishing magnetic property is the sign of the susceptibility χ_m: it is small and positive for paramagnetic materials (lines attracted) and small and negative for diamagnetic materials (lines expelled).

Final answer. Paramagnetic samples draw the field lines in ($\chi_m > 0$); diamagnetic samples expel them ($\chi_m < 0$).

← Back to Q51

Q52

The susceptibility of a magnetic material is -2.6×10^{-5}. *Identify the type of magnetic material and state its two properties.*

[2012]

SOLUTION

The magnetic susceptibility is given as $\chi_m = -2.6 \times 10^{-5}$, which is small in magnitude and negative in sign.

A small negative susceptibility is the characteristic signature of a diamagnetic material.

Two properties of a diamagnetic material:

1. Its susceptibility is small and negative ($\chi_m < 0$, relative permeability $\mu_r < 1$). The induced magnetisation $\vec{M}$ is directed opposite to the applied magnetising field, so the material is feebly magnetised against the field.

2. When placed in an external magnetic field, the field lines are expelled from the sample and bend around it. In a non-uniform field the substance moves from the stronger-field region to the weaker-field region, i.e. it is feebly repelled by a magnet, and its magnetic behaviour is independent of temperature.

Therefore the given material is **diamagnetic**.

Final answer. The material is diamagnetic (χ_m small and negative).

← Back to Q52

Q53

A short bar magnet of magnetic moment $0.9\ J/T$ *is placed with its axis at* $30°$ *to a uniform magnetic field. It experiences a torque of* $0.063\ J$.
(i) Calculate the magnitude of the magnetic field.
(ii) In which orientation will the bar magnet be in stable equilibrium in the magnetic field?

[2012]

SOLUTION

Given: magnetic moment $\boldsymbol{m = 0.9\ J/T}$, angle $\theta = 30°$, torque $\tau = 0.063\ J$ (i.e. 0.063 $N\ m$).

(i) A magnetic dipole of moment $\vec{m}$ in a uniform field $\vec{B}$ experiences a torque

$$\tau = mB\theta$$

$$\Rightarrow B = \frac{\tau}{m\theta}$$

$$B = \frac{0.063}{(0.9)(\sin 30°)}$$

$$B = \frac{0.063}{(0.9)(0.5)}$$

$$B = \frac{0.063}{0.45}$$

$$B = 0.14\ T$$

So the magnitude of the magnetic field is $\boldsymbol{0.14\ T}$.

(ii) The potential energy of the dipole is

$$U = -\vec{m}\cdot\vec{B} = -mB\theta$$

This energy is minimum (most negative) when $\theta = 1$, i.e. when $\theta = 0°$. A position of minimum potential energy is a position of stable equilibrium.

Hence the bar magnet is in **stable equilibrium when its magnetic moment** $\vec{m}$ **is parallel to** $\vec{B}$**, i.e. at** $\theta = 0°$ (the magnet aligned along the field).

Final answer. $B = 0.14\ T$; stable equilibrium occurs at $\theta = 0°$ ($\vec{m} \parallel \vec{B}$).

← *Back to Q53*

Q54

A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip down at 60° *with the horizontal. The horizontal component of the earth's magnetic field at the place is known to be* 0.4 *G. Determine the magnitude of the earth's magnetic field at the place.*

[2011 • Set 55-1-1]

SOLUTION

Given: angle of dip $\delta = 60°$, horizontal component $B_H = 0.4\ G$.

The horizontal component of the Earth's field is related to the total field B by

$$B_H = B\delta$$

$$\Rightarrow B = \frac{B_H}{\delta}$$

$$B = \frac{0.4}{\cos 60°}$$

$$B = \frac{0.4}{0.5}$$

$$B = 0.8\ G$$

Converting to SI units ($1\ G = 10^{-4}\ T$):

$$B = 0.8\times 10^{-4}\ T = 8\times 10^{-5}\ T$$

Hence the magnitude of the Earth's magnetic field at the place is $0.8\ G = 8\times 10^{-5}\ T$.

COMMON MISTAKE

The needle dipping at 60° *means the angle of dip is* $\delta = 60°$*; use* δ *for the horizontal component, not* δ.

Final answer. $B = 0.8\ G = 8\times 10^{-5}\ T$.

← Back to Q54

Q55

Explain the following:
(i) Why do magnetic lines of force form continuous closed loops?
(ii) Why are the field lines repelled (expelled) when a diamagnetic material is placed in an external uniform magnetic field?

[2011 • Set 55-2-1]

SOLUTION

(i) Why magnetic field lines form continuous closed loops:

Isolated magnetic poles (magnetic monopoles) do not exist; the simplest magnetic source is a dipole, where the north and south poles always occur together. As a result, magnetic field lines have no point at which they can start or stop.

This is expressed by Gauss's law for magnetism, which states that the net magnetic flux through any closed surface is zero:

$$\oint \vec{B}\cdot \mathrm{d}\vec{A} = 0$$

Since there are no sources or sinks of $\vec{B}$, every field line that leaves a region must re-enter it, so the lines must form continuous closed loops (they pass through the body of the magnet from S to N internally and from N to S externally).

(ii) Why field lines are expelled by a diamagnetic material:

A diamagnetic substance has atoms with no permanent magnetic dipole moment (all electron spins paired). When an external field $\vec{B}$ is applied, the orbital motion of the electrons changes, and by Lenz's law applied at the atomic level the induced current opposes the change in flux through each orbit.

The induced magnetic moment is therefore directed opposite to $\vec{B}$, producing a magnetisation $\vec{M}$ antiparallel to $\vec{B}$ (so $\chi_m<0$). This weakens the field inside the

sample, and the field lines are pushed out of, or repelled from, the diamagnetic body.

Final answer. Field lines close on themselves because $\oint \vec{B} \cdot d\vec{A} = 0$ (no monopoles); a diamagnet sets up $\vec{M}$ opposite to $\vec{B}$, so lines are expelled.

← *Back to Q55*

Q56

(i) Name the three elements of the Earth's magnetic field.
(ii) Where on the surface of the Earth is the vertical component of the Earth's magnetic field zero?

[2011 • Set 55-2-1]

SOLUTION

(i) The three elements of the Earth's magnetic field at a place are:

1. Magnetic declination (θ): the angle between the geographic meridian and the magnetic meridian at the place.

2. Angle of dip or magnetic inclination (δ): the angle which the Earth's total magnetic field $\vec{B}$ makes with the horizontal direction.

3. Horizontal component of the Earth's magnetic field (B_H): the component of $\vec{B}$ along the horizontal in the magnetic meridian.

Together these three elements completely specify the magnitude and direction of the Earth's magnetic field at that place.

(ii) The vertical component of the Earth's field is

$$B_V = B\delta$$

This vanishes when $\delta = 0°$, that is, where the field is purely horizontal.

This occurs at the **magnetic equator**. So the vertical component of the Earth's magnetic field is zero at the magnetic equator.

Final answer. The three elements are declination, angle of dip and horizontal component; $\mathbf{B_V=0}$ at the magnetic equator.

← *Back to Q56*

Q57

(i) Write two characteristics of a material used for making permanent magnets.
(ii) Why is core of an electromagnet made of ferromagnetic materials?

[2010]

SOLUTION

(i) Two characteristics of a material used for making permanent magnets:

1. High retentivity (high remanence): it should retain a strong residual magnetisation even after the magnetising field is removed, so that it stays a strong magnet.

2. High coercivity: it should require a large reverse magnetic field to be demagnetised, so that it does not lose its magnetism easily due to stray fields, mechanical shock or heat.

(Suitable materials are steel and alnico, which combine high retentivity with high coercivity.)

(ii) Why the core of an electromagnet is made of ferromagnetic material:

A ferromagnetic material such as soft iron has a very high relative permeability ($\mu_r \sim 10^3$ to 10^4). When used as the core, it greatly increases the magnetic field for a given current in the coil, making the electromagnet very strong.

Soft iron also has low retentivity and low coercivity, so it loses almost all of its magnetism the instant the current is switched off. This gives the electromagnet the essential property of being magnetic only while the current flows and non-magnetic when it stops.

COMMON MISTAKE

Permanent magnets need HIGH coercivity, but electromagnet (soft-iron) cores need LOW coercivity and low retentivity; do not mix the two requirements.

Final answer. Permanent magnets need high retentivity and high coercivity; electromagnet cores use ferromagnetic (soft iron) for high permeability with low retentivity.

← *Back to Q57*

Q58

Draw magnetic field lines when a (i) diamagnetic, (ii) paramagnetic substance is placed in an external magnetic field. Which magnetic property distinguishes this behaviour of the field lines due to the two substances?

[2010]

SOLUTION

Magnetic field lines in an external field $\vec{B}$:

(i) Diamagnetic substance: Draw a uniform horizontal field of equally spaced straight lines and an oval sample in the middle. The lines bend outward and avoid the sample, so the field inside is weaker than outside; some lines are expelled and bow around the body. The magnetisation $\vec{M}$ is opposite to $\vec{B}$.

(ii) Paramagnetic substance: Draw the same uniform field with an oval sample. The lines bend slightly inward and crowd together inside the sample, so the field inside is slightly stronger than outside. The magnetisation $\vec{M}$ is parallel to $\vec{B}$.

Magnetic property that distinguishes the two behaviours:

The behaviour is distinguished by the magnetic susceptibility χ_m (equivalently the relative permeability $\mu_r = 1+\chi_m$):

For a diamagnetic substance χ_m is small and negative ($\mu_r<1$), so the induced moment is against the field and the lines are expelled.

For a paramagnetic substance χ_m is small and positive ($\mu_r>1$), so the induced moment is along the field and the lines are concentrated.

Thus the **sign of the magnetic susceptibility χ_m** distinguishes the two responses.

Final answer. Diamagnetic samples expel the lines ($\chi_m<0$), paramagnetic samples concentrate them ($\chi_m>0$); the sign of χ_m is the distinguishing property.

← *Back to Q58*

Q59

Define magnetic susceptibility of a material. Name two elements, one having positive susceptibility and the other having negative susceptibility. What does negative susceptibility signify?

[2008]

SOLUTION

Magnetic susceptibility (χ_m):

It is defined as the ratio of the intensity of magnetisation $\vec{M}$ induced in a material to the magnetising field intensity $\vec{H}$:

$$\chi_m = \frac{M}{H}$$

It is a dimensionless number that measures how easily a material gets magnetised when placed in a magnetising field.

Two elements:

Positive susceptibility: Aluminium (a paramagnetic element, $\chi_m \sim +210^{-5}$).

Negative susceptibility: Copper (a diamagnetic element; bismuth is another example).

Significance of negative susceptibility:

A negative value of χ_m means the induced magnetisation $\vec{M}$ is in the direction opposite to the applied magnetising field $\vec{H}$. This is the characteristic of a **diamagnetic** material: it is feebly magnetised against the field, the field lines are expelled from it, and it is weakly repelled by a magnet.

Final answer. $\chi_m=M/H$; aluminium has positive χ_m, copper has negative χ_m; a negative value means $\vec{M}$ opposes $\vec{H}$ (diamagnetism).

← Back to Q59

Q60

Define the terms 'Magnetic Dip' and 'Magnetic Declination' with the help of relevant diagrams.

[2005]

SOLUTION

Magnetic Dip (angle of dip, δ):

It is the angle that the direction of the Earth's total magnetic field $\vec{B}$ makes with the horizontal at a given place.

Diagram: Draw a vertical plane (the magnetic meridian). Mark a horizontal reference line and the vertical direction. Draw the resultant field $\vec{B}$ tilted below the horizontal; the angle between $\vec{B}$ and the horizontal line is the dip δ. Show its components: $\mathbf{B_H = B\delta}$ along the horizontal and $\mathbf{B_V = B\delta}$ along the vertical, so that $\boldsymbol{\delta} = \dfrac{\mathbf{B_V}}{\mathbf{B_H}}$. (At the magnetic equator $\boldsymbol{\delta} = \mathbf{0}°$; at the magnetic poles $\boldsymbol{\delta} = \mathbf{90}°$.)

Magnetic Declination (θ):

It is the angle between the geographic meridian (the plane through geographic north-south) and the magnetic meridian (the vertical plane through the magnetic axis of a freely suspended needle) at a given place.

Diagram: Draw a horizontal, top-down view. Mark the geographic north-south line (true north) and the magnetic north-south line (the compass needle direction). The angle between these two lines, measured east or west of true north, is the declination θ.

Thus dip is measured in the vertical plane and declination in the horizontal plane.

Final answer. Dip $\boldsymbol{\delta}$ is the angle of $\vec{\boldsymbol{B}}$ with the horizontal; declination $\boldsymbol{\theta}$ is the angle between the geographic and magnetic meridians.

← Back to Q60

Q61

Explain with the help of diagram the terms (i) magnetic declination and (ii) angle of dip at a given place.

[2004]

SOLUTION

(i) Magnetic declination: At any place, the vertical plane containing the geographic north-south line is called the geographic meridian, and the vertical plane containing the magnetic axis of a freely suspended compass needle is called the magnetic meridian. The magnetic declination is defined as the angle between the geographic meridian and the magnetic meridian at that place.

In other words, it is the angle by which the compass needle (pointing to magnetic north) deviates from the true geographic north. It is usually denoted by θ and measured in degrees east or west of true north.

Diagram (top view of the horizontal plane): Draw a vertical straight line pointing up and label its tip N_g (geographic north). From the same starting point, draw a second line tilted by a small angle to one side (say towards the east) and label its tip N_m (magnetic north); this is the direction the compass needle settles in. The angle between these two lines, marked θ, is the magnetic declination.

(ii) Angle of dip (inclination): The angle of dip at a place is defined as the angle that the total magnetic field of the Earth $\vec{B}$ makes with the horizontal direction, in the magnetic meridian. It is denoted by δ.

If B_H is the horizontal component and B_V the vertical component of the Earth's field, then $\boldsymbol{\delta = \frac{B_V}{B_H}}$.

Diagram (vertical cross-section in the magnetic meridian): Draw a horizontal line through a point O to represent the horizontal direction, and label it B_H. From O, draw the resultant field vector $\vec{B}$ pointing into the ground at an angle below the horizontal. Drop a vertical line to complete a right-angled triangle; the vertical side is the vertical component B_V. The angle between $\vec{B}$ and the horizontal B_H, marked δ, is the angle of dip.

At the magnetic equator the field is horizontal, so $\delta = 0°$; at the magnetic poles the field is vertical, so $\delta = 90°$.

TIP

In the dip diagram, always show $\vec{B}$ below the horizontal with both components B_H and B_V labelled. The examiner gives the diagram mark only when the components are clearly shown.

Final answer. Declination is the angle between the geographic and magnetic meridians; dip is the angle δ the Earth's field makes with the horizontal, with $\delta = B_V/B_H$.

← *Back to Q61*

Q62

The magnetic moment (5 J/T) of a bar magnet points along a uniform magnetic field 0.4 T.
***(a)** Calculate (i) the potential energy of the bar magnet, and (ii) the work done in turning the magnet by 180°.*
***(b)** In which case is the potential energy of the magnet minimum?*

[2025 • Set 55-5-3]

SOLUTION

Given: magnetic moment $m = 5\ J/T$, uniform field $B = 0.4\ T$. Initially $\vec{m}$ points along $\vec{B}$, so the initial angle is $\theta_i = 0°$.

The potential energy of a magnetic dipole in a uniform field is $U = -\vec{m}\cdot\vec{B} = -mB\theta$.

(a)(i) Potential energy in the given (aligned) position:

$$U_i = -mB\theta_i$$

$$U_i = -(5)(0.4)\cos 0°$$

$$U_i = -(5)(0.4)(1)$$

$$U_i = -2.0\,J$$

(a)(ii) When the magnet is turned by 180°, the final angle is $\theta_f = 180°$:

$$U_f = -mB\cos 180°$$

$$U_f = -(5)(0.4)(-1)$$

$$U_f = +2.0\,J$$

The work done in turning the magnet equals the change in potential energy:

$$W = U_f - U_i$$

$$W = (+2.0) - (-2.0)$$

$$W = 4.0\,J$$

(b) Since $U = -mB\theta$, the potential energy is least (most negative) when $\theta = +1$, that is when $\theta = 0°$ and $\vec{m} \parallel \vec{B}$. This is the given initial orientation, where $U = -mB = -2.0\,J$. This is the position of **stable equilibrium**.

Final answer. $U_i = -2.0\,J$, work done $W = 4.0\,J$, and the potential energy is minimum when $\vec{m}$ is parallel to $\vec{B}$ (the initial position).

← *Back to Q62*

Q63

(a) A bar magnet of magnetic moment 2.5 JT^{-1} *lies aligned with the direction of a uniform magnetic field of* 0.32 *T.*
Find the amount of work done to turn the magnet so as to align its magnetic moment (i) normal to the field direction, and (ii) opposite to the field direction.
(b) What is the torque on the magnet in above cases (i) and (ii)?

[2024 • Set 55-1-2]

SOLUTION

Given: magnetic moment $m = 2.5\,J\,T^{-1}$, uniform field $B = 0.32\,T$. The magnet starts aligned with the field, so $\theta_i = 0°$.

(a) The potential energy of a dipole in a field is $U = -\mathbf{mB\theta}$, so the work done in turning it from θ_i to θ_f is:

$$W = U_f - U_i = mB(\theta_i - \theta_f)$$

(i) To turn the moment normal to the field, $\theta_f = 90°$:

$$W_1 = mB(\cos 0° - \cos 90°)$$

$$W_1 = (2.5)(0.32)(1 - 0)$$

$$W_1 = 0.80\,J$$

(ii) To turn the moment opposite to the field, $\theta_f = 180°$:

$$W_2 = mB(\cos 0° - \cos 180°)$$

$$W_2 = (2.5)(0.32)(1 - (-1))$$

$$W_2 = (2.5)(0.32)(2)$$

$$W_2 = 1.60\,J$$

(b) The torque on the dipole is $\tau = \mathbf{mB\theta}$.

(i) At $\theta = 90°$:

$$\tau_1 = mB\sin 90°$$

$$\tau_1 = (2.5)(0.32)(1)$$

$$\tau_1 = 0.80\,N\,m$$

(ii) At $\theta = 180°$:

$$\tau_2 = mB\sin 180°$$

$$\tau_2 = (2.5)(0.32)(0)$$

$$\tau_2 = 0$$

Final answer. Work done: $W_1 = 0.80\,J$ (normal) and $W_2 = 1.60\,J$ (opposite); torques: $\tau_1 = 0.80\,N\,m$ and $\tau_2 = 0$.

← Back to Q63

Q64

What are ferromagnetic materials? Explain ferromagnetism with the help of suitable diagrams, using the concept of magnetic domain.

[2024 • Set 55-4-1]

SOLUTION

Ferromagnetic materials are substances that are very strongly attracted by a magnet and can be permanently magnetised. For them the magnetic susceptibility is large and positive ($\chi_m \gg 1$) and the relative permeability is very high ($\mu_r \gg 1$). Examples are iron, cobalt, nickel, gadolinium and alloys such as steel and alnico.

Explanation using magnetic domains: A ferromagnetic material is divided into a large number of tiny regions called magnetic domains, each about 10^{-6} to 10^{-4} m across and containing billions of atoms. Inside a single domain, the atomic magnetic moments are all locked parallel to one another by a strong quantum-mechanical exchange interaction, so each domain is fully magnetised.

In an unmagnetised piece, the different domains point in random directions, so their magnetic moments cancel out and the net magnetisation is zero.

When an external magnetic field $\vec{B}$ is applied, magnetisation builds up in two ways. First, by domain wall motion: domains already favourably aligned with $\vec{B}$ grow in size at the cost of the others, as the boundaries (domain walls) shift. Second, by domain rotation: in strong fields the moments within each domain swing round to point along $\vec{B}$. The result is a large net magnetisation in the direction of the field.

When the external field is removed, many domains stay aligned, so the material retains magnetisation. This leftover magnetisation is called **remanence**, and it is the basis of permanent magnets.

Diagram description: Draw two square blocks. (a) Unmagnetised state: divide the block into several small cells by lines, and draw an arrow in each cell pointing in a different, random direction, so the arrows clearly do not agree,

giving zero net moment. (b) Magnetised state (field $\vec{B}$ shown by a horizontal arrow beside the block): draw the same block but now with most arrows aligned along $\vec{B}$, and show the favourably-aligned domains larger than the rest, illustrating domain growth and rotation.

Final answer. Ferromagnetic materials ($\chi_m \gg 1$, e.g. iron, cobalt, nickel) consist of randomly oriented domains; an applied field aligns them by domain wall motion and rotation, producing strong magnetisation that partly persists as remanence.

← *Back to Q64*

Q65

State Gauss's law for magnetism. Explain its significance.

[2019 • Set 55-1-1]

SOLUTION

Gauss's law for magnetism states that the net magnetic flux through any closed surface is always zero:

$$\oint \vec{B} \cdot \mathrm{d}\vec{S} = 0$$

Significance:

(1) It tells us that isolated magnetic monopoles do not exist. Unlike electric charges, a single north pole or a single south pole has never been found; magnetic poles always occur together as a north-south dipole. Even if a magnet is broken into pieces, each piece is again a complete dipole.

(2) It implies that magnetic field lines are always continuous closed loops. Since the total flux out of any closed surface is zero, every field line that leaves the surface must re-enter it; there is no point where a line simply begins or ends. This is in contrast with electric field lines, which start on positive charges and end on negative charges.

(3) Applied to a closed surface enclosing a bar magnet, the law shows that the inward flux (near the S-pole type lines) exactly equals the outward flux (near

the N-pole type lines), so the total magnetic flux is zero. This is consistent with the absence of a net magnetic charge inside the surface.

Final answer. Gauss's law for magnetism, $\oint \vec{B} \cdot d\vec{S} = 0$, signifies that magnetic monopoles do not exist and that magnetic field lines are continuous closed loops.

← *Back to Q65*

Q66

Write the four important properties of the magnetic field lines due to a bar magnet.

[2019 • Set 55-1-1]

SOLUTION

Four important properties of the magnetic field lines of a bar magnet:

(1) Continuous closed loops: The field lines are continuous closed curves. Outside the magnet they run from the north pole to the south pole, and inside the magnet they continue from the south pole back to the north pole, so each line is a complete loop. This is consistent with Gauss's law for magnetism, $\oint \vec{B} \cdot d\vec{S} = 0$.

(2) Tangent gives the field direction: The tangent drawn to a field line at any point gives the direction of the net magnetic field $\vec{B}$ at that point.

(3) Density indicates field strength: The number of field lines crossing a unit area (their crowding) represents the magnitude of $\vec{B}$. Where the lines are close together (as near the poles) the field is strong; where they are far apart the field is weak.

(4) Lines never intersect: Two magnetic field lines can never cross each other. If they did, there would be two different directions of $\vec{B}$ at the point of crossing, which is impossible because the field has a unique direction at every point.

Final answer. Magnetic field lines of a bar magnet form continuous closed loops, are tangent to $\vec{B}$ at each point, are denser where the field is stronger, and never intersect.

← Back to Q66

Q67

Write three points of differences between para-, dia- and ferro-magnetic materials, giving one example for each.

[2019 • Set 55-1-1]

SOLUTION

Three points of difference between diamagnetic, paramagnetic and ferromagnetic materials:

(1) Magnetic susceptibility χ_m: For diamagnetic materials it is small and negative ($\chi_m \approx -10^{-5}$). For paramagnetic materials it is small and positive ($\chi_m \approx +10^{-3}$). For ferromagnetic materials it is large and positive ($\chi_m \approx 10^3$ to 10^5).

(2) Relative permeability μ_r: For diamagnetic materials μ_r is slightly less than 1 ($\mu_r < 1$). For paramagnetic materials μ_r is slightly greater than 1 ($\mu_r > 1$). For ferromagnetic materials μ_r is very much greater than 1 ($\mu_r \gg 1$).

(3) Behaviour in an external magnetic field: A diamagnetic material is feebly repelled and tends to move from a stronger to a weaker field (field lines are pushed out). A paramagnetic material is feebly attracted and moves towards the stronger field (field lines slightly concentrate inside it). A ferromagnetic material is strongly attracted and strongly concentrates the field lines inside it.

Examples: Diamagnetic, copper (or bismuth, water); Paramagnetic, aluminium (or sodium); Ferromagnetic, iron (or cobalt, nickel).

TIP

Presenting these three differences as a neat three-column table earns full marks faster than prose and makes each contrasting point obvious to the examiner.

Final answer. Diamagnetic (e.g. copper): χ_m small negative, $\mu_r<1$, feebly repelled; Paramagnetic (e.g. aluminium): χ_m small positive, $\mu_r>1$, feebly attracted; Ferromagnetic (e.g. iron): χ_m large positive, μ_r1, strongly attracted.

← Back to Q67

Q68

(a) Show that the time period (T) of oscillations of a freely suspended magnetic dipole of magnetic moment (m) in a uniform magnetic field (B) is given by $T = 2\pi\sqrt{\frac{I}{mB}}$, *where I is a moment of inertia of the magnetic dipole.*
(b) Identify the following magnetic materials:
(i) A material having susceptibility (χ_m) = −0.00015.
(ii) A material having susceptibility (χ_m) = 10^{-5}.

[2019 • Set 55-3-1]

SOLUTION

(a) Consider a magnetic dipole of moment $\vec{m}$ freely suspended so that it can oscillate in a uniform field $\vec{B}$. Its equilibrium position is with $\vec{m}$ along $\vec{B}$.

When the dipole is displaced through a small angle θ from equilibrium, the field exerts a restoring torque on it:

$$\tau = -mB\theta$$

The negative sign shows the torque opposes the displacement. For small angles, $\theta \approx \theta$, so:

$$\tau = -mB\theta$$

By Newton's second law for rotation, with I the moment of inertia of the dipole:

$$I\frac{d^2\theta}{dt^2} = -mB\theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{mB}{I}\theta$$

This is the standard equation of angular simple harmonic motion, $\frac{d^2\theta}{dt^2} = -\omega^2\theta$, with:

$$\omega^2 = \frac{mB}{I}$$

The time period is $T = \frac{2\pi}{\omega}$, so:

$$T = 2\pi\sqrt{\frac{I}{mB}}$$

which is the required result.

(b) (i) $\chi_m = -0.00015 = -1.510^{-4}$ is small and negative, so the material is **diamagnetic**.

(ii) $\chi_m = 10^{-5}$ is small and positive, so the material is **paramagnetic**.

Final answer. Simple harmonic oscillation of the dipole gives $T = 2\pi\sqrt{I/(mB)}$; the material with $\chi_m = -1.510^{-4}$ is diamagnetic and the one with $\chi_m = 10^{-5}$ is paramagnetic.

← *Back to Q68*

Q69

Answer the following:
(a) Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why?
(b) Does a bar magnet exert a torque on itself due to its own field? Justify your answer.
(c) When an electron revolves around a nucleus, obtain the expression for the magnetic moment associated with it.

[2019 • Set 55-3-2]

SOLUTION

(a) A toroid is a closed ring of windings, so the magnetic field lines run round and round entirely inside the core. Being a closed structure, it has no ends from which the lines can escape, and applying Ampere's law to a loop lying outside the windings gives $\vec{B} = 0$ there. Hence the field lines are completely confined within the toroid.

A straight solenoid, on the other hand, has two open ends. Since the lines must close on themselves (because $\oint \vec{B} \cdot d\vec{S} = 0$), they have to come out at one end, spread through the space outside, and re-enter at the other end. So a stray field

exists outside a straight solenoid, especially near its ends, and the lines are not fully confined.

(b) No, a bar magnet does not exert a torque on itself due to its own field. A magnet is a rigid body, and by Newton's third law the forces between its own parts occur in equal and opposite pairs. These internal forces and the torques they produce cancel out, giving zero net force and zero net torque on the magnet as a whole. (If it could, the magnet would start rotating on its own, violating conservation of energy and angular momentum.)

(c) Consider an electron of charge e moving with speed v in a circular orbit of radius r around the nucleus. The time taken for one revolution is $\boldsymbol{T = \frac{2\pi r}{v}}$, so the orbiting electron is equivalent to a current:

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

The area enclosed by the orbit is $\boldsymbol{A = \pi r^2}$. The magnetic moment of a current loop is $\boldsymbol{\mu = IA}$, so:

$$\mu = \frac{ev}{2\pi r} \times \pi r^2$$

$$\mu = \frac{evr}{2}$$

This is the magnetic moment associated with the orbiting electron, directed perpendicular to the plane of the orbit.

Final answer. (a) A toroid is closed so its lines cannot leak out, unlike an open-ended solenoid; (b) no, a magnet exerts no net torque on itself; (c) $\boldsymbol{\mu = \frac{evr}{2}}$.

← *Back to Q69*

Q70

Define the dipole moment of a magnetic dipole. Write its S.I. unit.
Obtain the expression for the torque acting on a magnetic dipole placed in an external

uniform magnetic field.

[2019 • Set 55-3-3]

SOLUTION

Magnetic dipole moment: The magnetic dipole moment of a magnetic dipole is a vector that measures the strength and orientation of the dipole. For a bar magnet of pole strength q_m and magnetic length 2 (the distance between the poles), it is defined as:

$$\vec{m} = q_m\,(2)\,\hat{n}$$

where $\hat{n}$ is a unit vector directed from the south pole to the north pole of the magnet. (For a current loop of N turns, area A and current I, it is $\vec{\boldsymbol{m}} = \mathbf{NI}\vec{\boldsymbol{A}}$.)

S.I. unit: ampere metre^2 ($A\ m^2$), which is equivalent to joule per tesla ($J\ T^{-1}$).

Torque on a magnetic dipole in a uniform field: Consider a bar magnet of moment $\vec{m}$ placed in a uniform field $\vec{B}$, with its axis making an angle θ with the field. A force $+q_m\vec{B}$ acts on the north pole and an equal, opposite force $-q_m\vec{B}$ acts on the south pole. These two forces are equal in magnitude and opposite in direction, so they form a couple.

Diagram: Draw the magnet as a bar tilted at angle θ to a set of horizontal field lines $\vec{B}$. At the N-pole show a force q_mB along $\vec{B}$, and at the S-pole an equal force q_mB opposite to $\vec{B}$. The perpendicular distance between the lines of action of these two forces is 2θ.

The magnitude of the torque (force times perpendicular distance) is:

$$\tau = (q_m\,B)(2\theta)$$

$$\tau = (q_m \cdot 2)\,B\theta$$

$$\tau = mB\theta$$

In vector form this is:

$$\vec{\tau} = \vec{m} \times \vec{B}$$

This torque tends to align $\vec{m}$ with $\vec{B}$. It is maximum ($\boldsymbol{\tau = mB}$) when $\boldsymbol{\theta = 90°}$ and zero when $\boldsymbol{\theta = 0°}$ or 180°.

Final answer. The magnetic dipole moment $\vec{m} = q_m(2)\hat{n}$ has SI unit $A\ m^2$ ($= J\ T^{-1}$), and the torque on it in a uniform field is $\vec{\tau} = \vec{m} \times \vec{B}$, of magnitude $\tau = mB\theta$.

← Back to Q70

Q71

A bar magnet of magnetic moment 6 J/T is aligned at 60° with a uniform external magnetic field of 0.44 T. Calculate (a) the work done in turning the magnet to align its magnetic moment (i) normal to the magnetic field, (ii) opposite to the magnetic field, and (b) the torque on the magnet in the final orientation in case (ii).

[2018]

SOLUTION

When a bar magnet of magnetic moment m is rotated in a uniform magnetic field B from an angle θ_i to an angle θ_f with the field, the work done against the magnetic torque is stored as the change in potential energy of the magnet.

The work done is given by:

$$W = mB(\theta_i - \theta_f)$$

Here the data given are: $m = 6\ J/T$, $B = 0.44\ T$, and the initial orientation $\theta_i = 60°$.

(a)(i) To align the magnetic moment normal to the field, the final angle is $\theta_f = 90°$.

$$W_1 = mB(\cos 60° - \cos 90°)$$

$$W_1 = (6)(0.44)(0.5 - 0)$$

$$W_1 = (6)(0.44)(0.5)$$

$$W_1 = 1.32\ J$$

So the **work done to turn the magnet normal to the field is 1.32 J.**

(a)(ii) To align the magnetic moment opposite to the field, the final angle is $\theta_f = 180°$.

$$W_2 = mB(\cos 60^\circ - \cos 180^\circ)$$

$$W_2 = (6)(0.44)(0.5 - (-1))$$

$$W_2 = (6)(0.44)(1.5)$$

$$W_2 = 3.96\,J$$

So the **work done to turn the magnet opposite to the field is 3.96 *J*.**

(b) The torque on the magnet in any orientation is $\tau = mB\theta$. In the final orientation of case (ii), $\theta = 180^\circ$.

$$\tau = mB\sin 180^\circ$$

$$\tau = (6)(0.44)(0)$$

$$\tau = 0$$

The **torque is zero** because the magnetic moment is antiparallel to the field. This is the position of unstable equilibrium.

CONCEPT

When $\theta = 180^\circ$ the torque is zero but the potential energy is maximum, so this is unstable equilibrium, not stable. Do not confuse zero torque with stability.

Final answer. (a) $W_1 = 1.32\,J$ (normal), $W_2 = 3.96\,J$ (opposite); (b) torque $\tau = 0$.

← *Back to Q71*

Q72

(a) An iron ring of relative permeability μ_r has windings of insulated copper wire of n turns per metre. When the current in the windings is I, find the expression for the magnetic field in the ring.
(b) The susceptibility of a magnetic material is 0.9853. *Identify the type of magnetic material. Draw the modification of the field pattern on keeping a piece of this material in a uniform magnetic field.*

[2018]

SOLUTION

(a) The iron ring with windings of insulated wire acts as a toroid (closed solenoid). For a toroid with n turns per metre carrying a current I, Ampere's circuital law gives the magnetic field along the mean axis in free space (air core) as:

$$B_0 = \mu_0 \, n \, I$$

When the core is a magnetic material of relative permeability μ_r, the field inside is increased by the factor μ_r because the material gets magnetised and adds its own contribution. Replacing μ_0 by $\boldsymbol{\mu = \mu_0 \mu_r}$ gives the field inside the ring:

$$B = \mu_0 \, \mu_r \, n \, I$$

This is the required expression for the **magnetic field inside the iron ring.**

(b) The given susceptibility is $\boldsymbol{\chi_m = 0.9853}$, a small positive number (positive and of order unity, much less than the very large values of ferromagnets). A small positive susceptibility is the signature of a **paramagnetic material.**

The relative permeability is $\boldsymbol{\mu_r = 1 + \chi_m = 1 + 0.9853 = 1.9853}$, which is slightly greater than 1, again confirming a paramagnetic substance.

Modification of the field pattern: when a paramagnetic piece is placed in a uniform external magnetic field, the field lines are slightly drawn into the material, so the lines crowd together a little inside the sample. Describe the diagram as follows: draw evenly spaced horizontal field lines for the uniform external field; on placing the paramagnetic bar in this field, the lines near the bar converge slightly towards it and pass through it, with the line density inside the bar a little greater than outside. The field inside is therefore somewhat stronger than the applied field.

COMMON MISTAKE

The number 0.9853 is the susceptibility, not the permeability. Treat it as χ_m: small and positive means paramagnetic, small and negative means diamagnetic.

Final answer. (a) $\boldsymbol{B = \mu_0 \mu_r \, nI}$ inside the ring; (b) the material is paramagnetic, and field lines are slightly drawn into and concentrated within it.

← Back to Q72

Q73

(i) Mention two properties of soft iron due to which it is preferred for making an electromagnet.
(ii) State Gauss's law in magnetism. How is it different from Gauss's law in electrostatics and why?
OR
Derive an expression for the axial magnetic field of a finite solenoid of length 2l *and radius r carrying current I. Under what condition does the field become equivalent to that produced by a bar magnet?*

[2016]

SOLUTION

(i) Two properties of soft iron that make it suitable for an electromagnet:

First, soft iron has a very high relative permeability ($\mu_r \sim 10^3$ to 10^4), so for a given coil current it produces a very strong magnetic field.

Second, soft iron has low retentivity and low coercivity, so when the magnetising current is switched off it loses almost all of its magnetism. This lets the electromagnet be turned on and off on demand, which is exactly what an electromagnet must do.

(ii) Gauss's law in magnetism states that the net magnetic flux through any closed surface is always zero:

$$\oint \vec{B} \cdot d\vec{S} = 0$$

Gauss's law in electrostatics states that the net electric flux through a closed surface equals the charge enclosed divided by ε_0:

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\varepsilon_0}$$

How they differ and why: in electrostatics the right side can be non-zero because isolated electric charges (monopoles) exist, and electric field lines begin on positive charges and end on negative charges. In magnetism the right

side is always zero because isolated magnetic poles (magnetic monopoles) do not exist. Magnetic poles always occur in N-S pairs, so magnetic field lines form continuous closed loops with no starting or ending point. Hence the flux entering any closed surface equals the flux leaving it, and the net flux is always zero.

OR

Axial field of a finite solenoid of length 2l, radius r, with n turns per metre carrying current I:

Treat the solenoid as a stack of circular current loops. A small element of length dx at distance x from the centre contains n dx turns. The axial field at a point P far on the axis, at distance R from the centre, due to this element is:

$$dB = \frac{\mu_0 (n\,dx)\, I\, r^2}{2[(R-x)^2 + r^2]^{3/2}}$$

For a point far away ($R \gg r$ and $R \gg l$), we approximate $\mathbf{[(R-x)^2 + r^2]^{3/2} \approx R^3}$, so:

$$B = \int_{-l}^{+l} \frac{\mu_0\, n\, I\, r^2}{2R^3}\, dx$$

$$B = \frac{\mu_0\, n\, I\, r^2}{2R^3}\,(2l)$$

$$B = \frac{\mu_0\, n\, I\, r^2\, l}{R^3}$$

Now the total number of turns is $\boldsymbol{N = n(2l)}$, and the magnetic moment of the solenoid is $\boldsymbol{m = N\,I\,A = n(2l)\,I\,(\pi\, r^2)}$. Writing the field in terms of m:

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2m}{R^3}$$

This is exactly the expression for the axial (end-on) field of a bar magnet of magnetic moment m. So **the axial field of a finite solenoid is equivalent to that of a bar magnet** when the point lies far on the axis, that is, when the distance R is much greater than the length and radius of the solenoid.

Final answer. Soft iron has high permeability and low retentivity/coercivity; Gauss's law in magnetism gives zero net flux because magnetic monopoles do not exist. (OR) $\boldsymbol{B = \frac{\mu_0}{4\pi}\frac{2m}{R^3}}$, equivalent to a bar magnet when $\boldsymbol{R \gg l, r}$.

← Back to Q73

Q74

In what way is Gauss's law in magnetism different from that used in electrostatics? Explain briefly.
The Earth's magnetic field at the Equator is approximately 0.4 *G. Estimate the Earth's magnetic dipole moment. Given: Radius of the Earth = 6400 km.*

[2015]

SOLUTION

Gauss's law in magnetism states that the net magnetic flux through any closed surface is zero:

$$\oint \vec{B} \cdot \mathrm{d}\vec{S} = 0$$

Gauss's law in electrostatics states that the net electric flux through a closed surface equals the enclosed charge divided by ε_0:

$$\oint \vec{E} \cdot \mathrm{d}\vec{S} = \frac{q_{enc}}{\varepsilon_0}$$

The difference is that the electric flux can be non-zero because isolated electric charges exist, whereas the magnetic flux is always zero because isolated magnetic monopoles do not exist. Magnetic poles always come in N-S pairs, so magnetic field lines are continuous closed loops with no beginning or end. Electric field lines, on the other hand, begin on positive charges and end on negative charges.

Now estimate the Earth's magnetic dipole moment from the equatorial field. At an equatorial point of a dipole, the magnetic field is:

$$B_{eq} = \frac{\mu_0}{4\pi} \cdot \frac{m}{R^3}$$

Solving for the dipole moment m:

$$m = \frac{4\pi R^3 B_{eq}}{\mu_0} = \frac{R^3 B_{eq}}{10^{-7}}$$

since $\frac{\mu_0}{4\pi} = 10^{-7}\ T\ m/A$.

Substitute the data. The radius is $R = 6400\ \text{km} = 6.4\times 10^{6}\ m$, and the equatorial field is $B_{eq} = 0.4\ G = 0.4\times 10^{-4}\ T = 4\times 10^{-5}\ T$.

$$R^3 = (6.4\times 10^{6})^3 = 2.62\times 10^{20}\ \text{m}^3$$

$$m = \frac{(2.62\times 10^{20})(4\times 10^{-5})}{10^{-7}}$$

$$m = \frac{10.49\times 10^{15}}{10^{-7}}$$

$$m = 1.05\times 10^{23}\ A\ \text{m}^2$$

So the **Earth's magnetic dipole moment is about $1.05\times 10^{23}\ A\ \text{m}^2$.**

COMMON MISTAKE

Convert gauss to tesla before substituting: $1\ G = 10^{-4}\ T$. *Forgetting this factor of* 10^{-4} *is the most common error in this numerical.*

Final answer. Magnetism gives zero net flux (no monopoles), unlike electrostatics; Earth's magnetic dipole moment $m \approx 1.05\times 10^{23}\ A\ \text{m}^2$.

← Back to Q74

Q75

What are permanent magnets? What is an efficient way of preparing a permanent magnet? Write two characteristic properties of materials which are required to select them for permanent magnets.

[2008]

SOLUTION

Permanent magnets are materials that retain a strong magnetisation even after the external magnetising field is removed. They keep their magnetism through ordinary handling and use because they have high retentivity and high coercivity.

Efficient way of preparing a permanent magnet: place the ferromagnetic specimen, such as a steel rod, inside a long solenoid and pass a strong direct current through the solenoid. The strong magnetic field produced inside the solenoid aligns the magnetic domains of the specimen along the field. When the current is switched off, the steel keeps this domain alignment because of its high retentivity, and so it becomes a permanent magnet. This solenoid method gives a stronger and more uniform magnetisation than the older single-touch or double-touch stroking methods.

Two characteristic properties required for selecting a material for permanent magnets:

First, high retentivity, so that the material retains a large residual magnetisation when the magnetising field is removed.

Second, high coercivity, so that the material is not easily demagnetised by stray external fields, mechanical shocks or temperature changes, and stays magnetic over a long working life.

Steel and alloys such as Alnico are common materials for permanent magnets.

Final answer. Permanent magnets retain magnetisation after the field is removed; prepared by placing the specimen in a solenoid carrying a strong DC current; the material must have high retentivity and high coercivity.

← *Back to Q75*

Q76

Distinguish between diamagnetic and ferromagnetic materials in respect of their (i) intensity of magnetisation, (ii) behaviour in a non-uniform magnetic field and (iii) susceptibility.

[2003]

SOLUTION

The three required differences between diamagnetic and ferromagnetic materials are best presented in a table.

(i) Intensity of magnetisation M: In a diamagnetic material, M is small and negative, that is, it is directed opposite to the applied field $\vec{B}$, and its magnitude is very small. In a ferromagnetic material, M is large and positive, that is, directed along $\vec{B}$, and its magnitude is very large.

(ii) Behaviour in a non-uniform magnetic field: A diamagnetic substance moves from the stronger to the weaker region of the field, so it is weakly repelled by the magnet. A ferromagnetic substance moves from the weaker to the stronger region, so it is strongly attracted towards the magnet.

(iii) Susceptibility χ_m: For a diamagnetic material, χ_m is small and negative, of the order of -10^{-5}. For a ferromagnetic material, χ_m is large and positive, of the order of 10^3 to 10^5.

As examples, copper and bismuth are diamagnetic, while iron, nickel and cobalt are ferromagnetic.

Final answer. Diamagnetic: $\boldsymbol{M}$ small and negative, repelled from strong field, $\boldsymbol{\chi_m}$ small and negative; Ferromagnetic: $\boldsymbol{M}$ large and positive, attracted to strong field, $\boldsymbol{\chi_m}$ large and positive.

← *Back to Q76*

Q77

Define the terms: 'magnetic field strength', 'magnetic flux density' and 'permeability' of a magnetic material. Write their SI units.

[2010]

SOLUTION

Magnetic field strength (magnetising field) $\vec{H}$: it is the auxiliary magnetic field that measures the magnetising influence of free (external) currents alone, without the response of the material. In a material it is defined by:

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$

and in vacuum it reduces to $\vec{H} = \frac{\vec{B}}{\mu_0}$. Its SI unit is **ampere per metre (A/m).**

Magnetic flux density (magnetic field) $\vec{B}$: it is the total magnetic field at a point, accounting for both the external magnetising field and the magnetisation of the material. It is related to $\vec{H}$ and the magnetisation $\vec{M}$ by:

$$\vec{B} = \mu_0(\vec{H} + \vec{M})$$

Equivalently $\vec{B} = \mu\vec{H}$. The magnetic flux density also equals the magnetic flux per unit area held normal to the field. Its SI unit is the **tesla (T)**, where $\mathbf{1\ T = 1\ Wb/m^2}$.

Permeability of a magnetic material μ: it is the ratio of the magnetic flux density to the magnetising field in the material, that is:

$$\mu = \frac{B}{H}$$

It measures how easily a material allows magnetic field lines to pass through it, or how much it gets magnetised for a given magnetising field. It can also be written as $\mu = \mu_0\mu_r = \mu_0(1 + \chi_m)$, where μ_r is the relative permeability and χ_m the susceptibility. Its SI unit is **henry per metre (H/m)**, equivalently $T\ m/A$.

For free space the permeability is $\mu_0 = 4\pi \times 10^{-7}\ T\ m/A$.

TIP

Keep the units straight: $\vec{H}$ and $\vec{M}$ are both in A/m, while $\vec{B}$ is in tesla. Mixing the unit of H with that of B costs marks in unit-based questions.

Final answer. $\vec{H}$ (A/m) is the magnetising field, $\vec{B}$ (tesla) is the total flux density, and $\mu = B/H$ (H/m) is the permeability.

← *Back to Q77*

Q78

Distinguish the magnetic properties of dia-, para- and ferro-magnetic substances in terms of (i) susceptibility, (ii) magnetic permeability and (iii) coercivity. Give one example of each of these materials.

Draw the field lines due to an external magnetic field near a (i) diamagnetic, (ii) paramagnetic substance.

[2007]

SOLUTION

The magnetic properties of diamagnetic, paramagnetic and ferromagnetic substances are distinguished below in terms of susceptibility, permeability and coercivity, with one example of each.

(i) Susceptibility χ_m: For a diamagnetic substance, χ_m is small and negative, of the order of -10^{-5}. For a paramagnetic substance, χ_m is small and positive, of the order of $+10^{-5}$ to $+10^{-3}$. For a ferromagnetic substance, χ_m is large and positive, of the order of 10^3 to 10^5.

(ii) Magnetic permeability μ: For a diamagnetic substance, μ is slightly less than μ_0 (so $\boldsymbol{\mu_r < 1}$). For a paramagnetic substance, μ is slightly greater than μ_0 (so μ_r is slightly more than 1). For a ferromagnetic substance, μ is much greater than μ_0 (so $\mu_r \gg 1$).

(iii) Coercivity: For diamagnetic and paramagnetic substances coercivity is not applicable, since they retain no magnetisation when the field is removed. For a ferromagnetic substance coercivity is significant: it is low for soft ferromagnets such as soft iron and high for hard ferromagnets such as steel.

Examples: copper or bismuth is diamagnetic, aluminium or calcium is paramagnetic, and iron, nickel or cobalt is ferromagnetic.

Field lines near a diamagnetic substance: draw evenly spaced horizontal external field lines. On placing the diamagnetic sample, the field lines bend away and are expelled from the interior, so they bow around the sample and the line density inside the sample is less than outside. The diamagnetic body tends to move out of the field.

Field lines near a paramagnetic substance: draw evenly spaced horizontal external field lines. On placing the paramagnetic sample, the field lines are slightly drawn into the material, so they converge a little towards the sample and the line density inside is slightly greater than outside. The paramagnetic body tends to move into the field.

Final answer. Diamagnetic: χ_m small negative, $\mu<\mu_0$, no coercivity (e.g. Cu); paramagnetic: χ_m small positive, $\mu>\mu_0$, no coercivity (e.g. Al); ferromagnetic: χ_m large positive, $\mu\mu_0$, significant coercivity (e.g. Fe). Field lines are expelled from a diamagnetic and drawn into a paramagnetic sample.

← *Back to Q78*